



A two-stage risk mitigation-aware energy scheduling for integrated water-power system with hybrid energy convertors

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ABSTRACT

The increasing complexity and interconnectedness of Integrated Water-Power Systems (IWPS) are increasingly becoming riskability against unauthorized intrusions. This complexity requires innovative solutions to ensure system resilience, operational efficiency, and risk mitigation. This paper introduces a novel two-stage approach to managing these intricate IWPSs by integrating sensitivity analysis, risk assessment, and optimization methodologies. In this regard, first the key decision variables are identified by sensitivity analyzing and their impact on the system state variables is evaluated. To evaluate risks, particularly in the context of cyber threats such as False Data Injection Attacks (FDIA), a Proposed Jacobian Matrix (PJM) analysis is then introduced. Additionally, this paper offers a hybrid two-stage structure aimed at enhancing risk management in IWPS drawing on insights from the Proposed Jacobian Matrix (PJM) analysis. To overcome the IWPS's vulnerabilities and enhance optimal performance across operating conditions, the framework employed a multi-layered approach utilizing three key energy converters: water-to-thermal conversion, electricity-to-thermal conversion, and photovoltaic system within IWPS. Meanwhile, the first stage begins to analyze the hybrid structure of energy converters with a focus on risk minimization, specifically aimed at reducing the IWPS's vulnerability to potential disruptions and cyber threats. The second stage is defined based on a cost-effective operational strategy aimed at optimizing the decision variables across the hybrid structure applied in the first stage. The proposed framework is carried out in a studied IWPS with 36 electrical buses, 5 thermal buses, and 5 water buses across an interconnected network. The maximum estimated economic impact is 1.6×10^{13} , indicating that the threat from Foreign Data Injection Attacks (FDIA) to the electrical network is the highest, with a risk rating of 3.89. In comparison, the gas network has a risk rating of 3.58, and the water network has a rating of 3.64. Comparative studies show that the PJM technique provides more accurate and consistent risk assessments than Conditional Value-at-Risk (CVaR). Furthermore, the proposed architecture ensures a steady reduction in overall operating expenses while decreasing the Integrated Water and Power Supply (IWPS) risk coefficient from 5.45 to 3.17.

Nomenclature

Sets/Indices	
Ω^t/t	: Set/index of time, where $\Omega^t = \{1, \dots, 24\}$.
Ω^e/e	: Set/index of electrical bus.
Ω^{ww}/ww	: Set/index of water bus.
Ω^{RS}/RS	: Set/index of thermal bus.
Ω^l/l	: Set/index of power line.
Ω^{RL}/RL	: Set/index of thermal line.
Ω^{wl}/wl	: Set/index of water line.
Constants	
$p_{e,t}^{elec-load}$	Power load (kW)

(continued)

gb_l, bg_l	Technical characters of power line (p.u.)
$\underline{C}^{transfor}, \bar{C}^{transfor}$	Limitations of the Power injection from upstream network (kW)
$\underline{p}_{e,t}^{dichr}, \underline{p}_{e,t}^{chr}, \bar{p}_{e,t}^{dichr}, \bar{p}_{e,t}^{chr}$	Limitations of the storage unit (kW)
$H_{RS,t}^{thermal-load}$	Thermal load (kW)
C_{RL}	Temperature conversion coefficient of thermal line
$C_p, \Delta T_{gas}, C^{fuel}$	Transmission coefficients of thermal line
$\alpha_q, \eta_{chp}, \eta_{boi}$	Efficiency coefficients of CHP and boiler units
K_1, K_2, K_3	energy conversion coefficients

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$\bar{C}_{chp-heat}, \bar{C}_{chp-heat}^{Boiler}, \bar{C}_{Boiler}$	Limitations of CHP and boiler units (kW)
$W_{ww,t}^{water-load}$	Water load (m ³ /h)
W_{wl}^{base}	Base limitation of line water (m ³ /h)
$S, \Delta t$	constant coefficients of water pump (m ³ /h)
A_{wl}, B_{wl}, C_{wl}	constant coefficients
P, g, η_{des}	gravitational coefficient, the water density and the constant pump efficiency.
$\bar{C}_{desalination}$	Limitations of the desalination unit
β, η_W	energy conversion coefficients of water to thermal convertor
$\bar{C}_{water}, E_t^{H-storage}, E_t^{H-storage}$	Limitations of water to thermal convertor and storage unit ((m ³ /h), kW)
w, c_1, c_2	Setting parameters of PSO
X	Input data
e	Error parameter
$\bar{A}IL$	the Attack Injection Level
E, C^{PV}, g	Energy capacity, Solar radiation, and Direct irradiation
$X_{\kappa}^{mesu}, \sigma_{\kappa}^2, I_{\kappa}$	Measured data, variance, mean values
$loss^{PV}$	Power loss of PV (kW)
Variables	
$p_{e,t}^{transfor-up}$	The Power injection from upstream network (kW)
$p_{e,t}^{elec-line}$	Power injection at node (e) (kW)
$p_{e,t}^{chp-heat}$	Power output of CHP (kW)
$p_{e,t}^{battery}$	Power output of storage (kW)
$p_{e,t}^{desalination}$	Power output of desalination unit (kW)
V_e, δ_e	Voltage and angle of power bus (p.u., Deg.)
$p_{e,t}^{dichr}, p_{e,t}^{chr}, E_{e,t}^{battery}$	Discharge power, charge power, and energy of storage unit (kW)
$H_{RS,t}^{Boiler}$	Thermal output of boiler unit (kW)
$H_{RL(RS,RS),t}^{thermal-line}$	Thermal transmission in lines (kW)
$H_{RS,t}^{chp-heat}$	Thermal output of CHP unit (kW)
T_{RL}^S, T_{RL}^R	Line temperatures (C°)
$p_{gas-line}$	Gas consumption for line (kW)
p_q^{input}	output net power of CHP (kW)
$p_{gas-chp}$	Gas consumption for CHP (kW)
H_q^{input}	output net thermal of CHP (kW)
$p_{gas-boiler}$	Gas consumption for boiler (kW)
$p_{gas-total}$	Gas consumption for line (kW)
$W_{ww,t}^{grid-up}$	The water injection from upstream network (m ³ /h)
$W_{ww,t}^{wat-line}$	water transmission in lines (m ³ /h)
$h_{ww,t}$	Line Hydraulic (m ³ /h)
W_{water}	Water consumption of converter (m ³ /h)
$H_{W^{heat}}$	Thermal output of converter (m ³ /h)
$E_t^{H-storage}, H_t^{chr}, H_t^{dichr}$	Thermal, charge, and discharge thermals of storage (m ³ /h)
p^c, X_{κ}^i, p_g^c	Local Beat, input, and total beat location
z	Output value of FDIA
JM_{ij}	Elements of Jacobian matrix
PV	Power output of PV (kW)

1. Introduction

1.1. The motivation of this work

Integrated Energy Systems (IESs) have evolved to be more flexible, intelligent, and efficient with an emphasis on the synergism among various energy resources. In light of this, energy converters have attained a crucial role in the development of IESs in recent years. On the flip side, the high integration of energy convertors can naturally raise threats to IESs [1]. Due to their increasing interconnectedness, dependence on digital control systems, and operational complexity, Integrated Energy Systems (IESs) are becoming prime targets for sophisticated cyberattacks. One significant category of these cyber threats is False Data Injection Attacks (FDIAs), where an attacker falsifies measurement data to mislead control actions and degrade system performance. As energy systems evolve to include water infrastructures alongside electricity and heat, leading to the development of Integrated Water-Power

Systems (IWPS), the number of control points and converter-based interactions rises dramatically. This expansion increases the risk of cross-domain failures, as compromised components can trigger cascading effects throughout the interconnected subsystems, resulting in greater cyber exposure. While numerous studies have addressed cyber-physical risks in dual-network IESs, tri-network systems like IWPS have received less attention. There is a particular need to investigate the vulnerabilities posed by energy converters, which can serve as potential entry points and vectors for attacks. However, there is a pronounced lack of research in this area within the three-network system. Meanwhile, this paper tries to address an efficient risk assessment model against cyber-attacks for IESs including the water, electricity, and heat networks. Also, establishing a coordination framework to mitigate energy supply risks in IESs is another goal of this paper.

1.2. Literature review

IES enhances efficiency with the use of optimizing energy generation, distribution, and consumption. A more sustainable energy infrastructure is also linked to increased resilience and reliability. In Ref. [2], a model for effective IES energy management that utilizes both heat and electrical energy is proposed. To address the unpredictability of power prices and renewable energy, the authors in Ref. [3] explore a robust operational model for integrated electricity and heat systems (IEHS). Since IESs encompass multiple energy networks, they are particularly vulnerable to cyberattacks.

1) Cyber-attack in IESs

A cyberattack on any of the networks could impact the administration of the entire system, as these systems are monitored and operated in an integrated manner. In the following sections, we will examine several recent cyber incidents. The number of breakdowns in energy systems due to cyberattacks has significantly increased in recent years. One of the most notable incidents was the Colonial Pipeline ransomware attack in May 2021, which disrupted the supply chain and became one of the largest ransomware attacks in U.S. history [4]. Another major attack occurred during a continental energy crisis, targeting the Amsterdam-Rotterdam-Antwerp (ARA) region, a key hub for oil refining in Europe, thereby disrupting the loading and unloading of refined products [5]. Additionally, one of the worst hacks on the U.S. government took place in December 2020, affecting the Department of Energy [6]. Due to varying security and privacy regulations across different nations, many cyber-related incidents remain unreported. Therefore, it is essential to investigate the impact of cyber threats on IES's energy management, particularly the ripple effects that cyberattacks can have as they transfer from one energy system to another. Cyberattacks can significantly affect communications, control systems, and energy management. Common types of attacks on IES include denial of service (DoS), man-in-the-middle attacks, and FDI [7]. The existing literature has focused on modeling these attacks and establishing detection frameworks. Reference [8] provides a comprehensive overview of various techniques for identifying FDIAs in smart grids. Additionally, Reference [9] examines the impact of both static and dynamic FDI attacks on the electricity system, considering three key factors: control, state estimation, and economic dispatch. A cyberattack targeting one network (for instance: an electricity network) in IES can compromise the security of another, and failures in one system can trigger cascading issues in others. While interdependency can lead to several positive outcomes, such as enhanced supply security and greater utilization of renewable energy, it also has its drawbacks.

2) Risk assessment methods for IESs

According to IES, risk is defined as the measure of both the likelihood and severity of an undesirable outcome. In simpler terms, it assesses

safety by evaluating questions such as “How severe could an accident be?” and “How often might it occur?” Random equipment failures often occur outside the control of IES personnel, and the involvement of loads and renewable energy sources introduces inherent uncertainties that make precise forecasting impossible. Consequently, the effects of these fluctuations can vary, leading to localized energy interruptions or even the potential for widespread blackouts [10]. State selection methods, such as State Enumeration (SE) and Monte Carlo (MC) simulation [11], are commonly used to assess the risk evaluation of IES. Several efforts have been made to conduct contingency analyses of the consequences for the entire system, aiming to improve risk assessment methodologies. This approach focuses on the coordinated post-contingency dispatch of electric-heat integrated energy systems, providing a system-level perspective on risk for independent energy operators [12]. Given the unpredictability of solar and wind energy, a data-driven approach is proposed to evaluate the likelihood of supply shortages in renewable energy systems that combine electricity and heat [13]. Additionally, while previous studies [14] highlight that reserving adequate line-pack space in Natural Gas Storage (NGS) can enhance energy capacity and mitigate the risk of shortages, they often overlook the failures of random components. Reference [15] emphasizes the need for a compressor station that incorporates multiple compressor units for the risk assessment of electric-gas IES. Reference [16] explores the use of the Gram-Charlier series as an effective method for evaluating the impact of power-to-gas (P2G) on operational risk in IES. Reference [17] analyzes the effects of solar thermal combined heat and power (CHP) systems and electric boilers on risk within IES, which encompasses gas, heat, and electric networks. This assessment considers component failures as well as the unpredictability of demand and renewable energy sources. Finally, reference [18] classifies risk situations into four categories: material flow, energy flow, energy-material flow coupling, and process feedback. This reference offers a more comprehensive modeling approach to risk assessment [10]. A multi-weighted risk grading model has been developed to assess the overall risk status of the system. To enhance the safety of the RIES, an emergency dispatch plan based on dynamic optimal energy flow is proposed. This plan considers the dynamic properties of both the gas and heat systems in the RIES risk assessment [19,20]. Additionally, a dynamic load restoration model has been created that connects gas and electric energy to minimize load loss. Unlike traditional methods that rely on annual average indices, this model can evaluate the risk indices of the RIES system for specific time periods. Furthermore, several studies have investigated the impact of cyberattacks on RIES risks, particularly given the increasing integration of IES and cyber-physical systems.

1.3. Research Gap and aim

As illustrated in Table 1, in the literature, several studies have only employed robust optimization techniques to mitigate energy risks in the operation of integrated heat and electrical networks. There are fewer studies that explore operational risk assessment in a three-network system while considering cyber-attacks. In such a three-network system, the types of energy converters can significantly affect the system vulnerability and increase the risk due to their dual role as producers in one network and consumers in another. In the event of a cyberattack, these converters may also be easily identified and targeted, which can alter system performance and functionality. Integrated Water-Power Systems (IWPS) combining three networks electrical, thermal, and water are inherently more complex and require additional energy converters. This issue poses a more significant and serious risk assessment model. Most research on risk assessment generally overlooked energy converters as potential targets for attackers within three-network systems. Consequently, it is essential to develop risk-reduction plans for cyber-attack-based high-risk scenarios. This research focuses on the role of energy converters in the context of cyberattack scenarios, presenting a novel two-stage operational risk management methodology for IWPS.

Table 1

The overview of risk assessment methods and analysis models for multi-energy networks in literature works.

References	Energy Network	Analysis model	Risk assessment method	Key consideration
[11]	Integrated energy system	state selection technique	Monte Carlo	contingency analyses of the consequences independent energy operators
[12]	Electric-thermal system	Coordinated post-contingency dispatch	State Enumeration and Monte Carlo	
[13]	Regional Electric thermal system	Multi-energy flow model	Data-driven	Renewable energy
[14]	–	Minimum operation cost model	Optimal method	overlook the failures of random components
[15]	Integrated energy system	Load shedding	Multi-level Monte Carlo	multiple compressor units
[16]	–	Multi-energy flow model	State Enumeration	Semi-invariant method
[17]	Renewable IES	Renewable and load curtailment method	Monte Carlo	solar thermal CHP
[18]	Electric-heat gas system	energy-material flow	multi-weighted risk grading model	dynamic optimal energy flow
[19]	Multi-energy system	MILP Model	Cvar Method	Voltage/var Control
Proposed	Integrated Water-Power System	Multi-energy flow model	Proposed Jacobian Matrix (PJM) analysis	Cyberattacks

The first step is to identify how FDIA affect various subsystems and to quantify the cyber-physical risks associated with converter components using an analysis based on the Proposed Jacobian Matrix (PJM). In the second step, an efficient and cost-effective operational strategy is developed based on the risk-constrained operating space established in the first step. This framework addresses a significant gap in the literature by providing a more reliable and practical approach to operating cyber-aware IWPS, explicitly modeling energy converters as both functionally critical and vulnerable components. This study introduces a framework for operational risk assessment and mitigation that takes into account the propagation of failures arising from FDIAs across systems within an IWPS. The primary contributions of this research are as follows:

Research Gap 1.

- Most current research lacks comprehensive operational risk assessment frameworks for tri-network systems (electricity, heat, and water), particularly in the context of cyberattacks. Instead, many studies focus on risk reduction in integrated heat and electrical networks through robust optimization.

Contribution 1.

- A new and precise method for assessing hazards in IWPS, including cyber threats such as FDIA, has been introduced using the Proposed Jacobian Matrix (PJM) analysis. The PJM method exhibits greater accuracy and consistency than traditional risk measures like Conditional Value-at-Risk (CVaR).

Research Gap 2.

- Traditional risk indicators, such as Conditional Value-at-Risk (CVaR), are commonly employed to assess cyber risks, especially in relation to False Data Injection Attacks (FDIA). However, these indicators often fail to adequately capture the dynamic impact of cyber threats within interconnected systems.

Contribution 2.

- Providing a hybrid two-stage framework for risk mitigation: In stage 1, the combination of energy converters (photovoltaic, electricity-to-thermal, and water-to-thermal systems) is analyzed and optimized to reduce risk and enhance resilience against disruptions and cyber-attacks. To maintain optimal system performance, stage 2 enhances stage 1 by implementing a cost-effective optimization of decision variables within the hybrid structure.

Research Gap 3.

- Additionally, research has largely overlooked the dual roles of energy converters as both risk propagators and vulnerable components in multi-network systems.

Contribution 3.

- Energy converter integration in multi-network systems: To effectively address vulnerabilities in IWPS, the framework incorporates three essential energy converters: solar systems, electricity-to-thermal conversion, and water-to-thermal conversion. This multi-layered approach ensures robust energy management under various operating conditions, enhancing overall operational performance.

1.4. Organization

The following sections are written: Section II: provides the problem

formulation to define the integrated water-power system model. Section III: the attack model adapted to IWPS is modeled in this section. Section IV: This section aims to present the risk assessment for IWPS. Section V: the coordinated probability-aware risk model for IWPS is presented. Section VI: Emulation and evaluation of the case study. Finality, Section V: is related to the conclusion.

2. The integrated water-power system model

To understand the challenges faced by the IES, it's essential to examine its basic structure. This section focuses on the mathematical model of the IWPS, which incorporates the upstream electrical grid. The primary goal of the IWPS discussed in this study is to efficiently meet energy demands by integrating three distinct energy networks: 1) the electricity grid, 2) the thermal network, and 3) the water network. Fig. 1 illustrates an overall view of the studied IWPS structure. Importantly, the inclusion of the upstream grids ensures the reliability of the IWPS. This section will primarily cover the IWPS.

1) Realistic Electricity Network Modelling

The methodology can be applied to unbalanced multiphase models; however, for simplicity, we will use a single-phase equivalent model for a balanced power network. The power network is represented as a directed graph with nodes (e). In such a network, energy management aims to satisfy power loads using energy resources. In this way, the power balance can be defined based on equation (1):

- Power Balance Equation

$$P_{(e,t)}^{(transfor-up)} + P_{(l(e,t),t)}^{(elec-line)} + P_{(e' \subseteq e,t)}^{(chp-heat)} + P_{(e,t)}^{battery} = P_{(e,t)}^{(elec-load)} + P_{(e' \subseteq e,t)}^{desalination} \quad (1)$$

Power flow: Let $P_{l(e,t),t}^{elec-line}$ represents the complex power injection at node (e) in line (l) at time (t). The power flow is calculated regarding the angle (δ_e) and voltage (V_e) of each node as shown in equation (2).

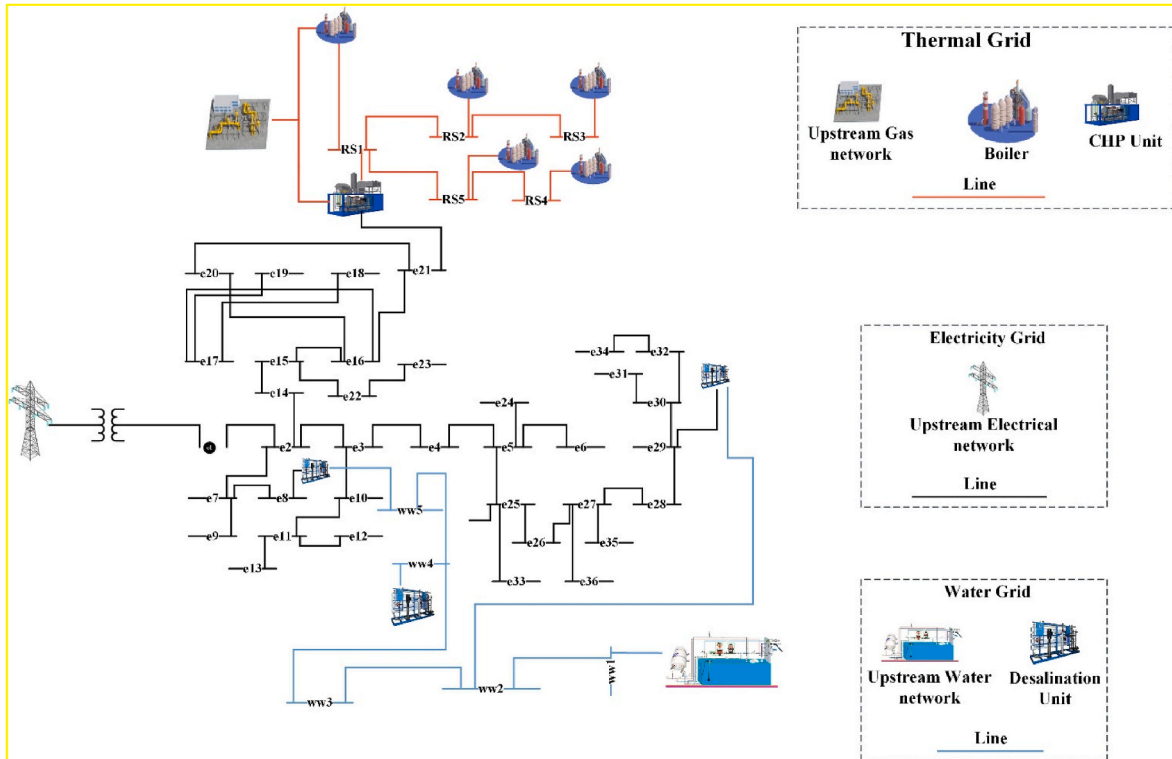


Fig. 1. Overview illustration of the proposed IWPS.

$P_{e,t}^{elec-load}$ models the load demands of the electricity network which should be satisfied at time t [31].

$$P_{e,e1}^{elec-line} = V_e^2 g b_l - V_e V_{e1} g b_l \cos(\delta_e) - V_e V_{e1} b g_l \sin(\delta_e) \quad (2)$$

- External electricity resources

$P_{e,t}^{transfor-up}$ models the delivering power from the upstream grid through the transformer unit. This is defined as an input generation resource for the electricity network. To limit power generation to its nominal capacity, equation (4) models the constraints related to power transactions in IWPS at time t more clearly.

$$P_{e,t}^{transfor-up} = \text{upstream grid} \quad (3)$$

$$\underline{C}^{transfor} \leq P_{e,t}^{transfor-up} \leq \overline{C}^{transfor} \quad (4)$$

- Electricity storage unit

The output of the storage unit is defined by the variable $P_{e,t}^{battery}$. Equation (5) defines the energy level of the storage at time t , while Equations (6)–(8) regulate the charging and discharging capacities of the system [32].

$$E_{e,t}^{battery} = E_{e,t-1}^{battery} + P_{e,t}^{chr} - P_{e,t}^{dichr} \quad (5)$$

$$P_{e,t}^{battery} = P_{e,t}^{dichr} - P_{e,t}^{chr} \quad (6)$$

$$P_{e,t}^{dichr}, P_{e,t}^{chr} \leq \overline{P}_{e,t}^{dichr}, \overline{P}_{e,t}^{chr} \leq \underline{P}_{e,t}^{dichr}, \underline{P}_{e,t}^{chr} \quad (7)$$

$$E_{e,t}^{battery} \leq \overline{E}_{e,t}^{battery} \leq \underline{E}_{e,t}^{battery} \quad (8)$$

- Electricity energy convertors

The power flow calculation requires the assumption of various energy convertors within IWPS [21]. For instance, let's consider a CHP unit that simultaneously generates electrical and thermal energy by converting input gas within the IWPS. In this context, $P_{e',t}^{chp-heat}$ will represent the power generation of the CHP unit in the power network, which we will elaborate on in the following sections. Additionally, a water desalination unit is utilized to convert seawater into consumable water. This unit requires power ($P_{e',t}^{desalination}$) that is accounted for as load demand within the power network.

2) Realistic Thermal Network Modelling

A typical thermal network structure consists of one source node and several load nodes connected by supply and return pipelines. The primary network includes the supply pipelines that transport thermal energy to the load nodes using water. This thermalized water then enters the thermal exchanger of the secondary network, resulting in a significant change in temperature. To create our model, we integrated data from sources [22–25] [main3], and [26]. The thermalizing network is represented as a directed graph ($RS; RL$), where RS is the node and RL is the pipe. Each node in the system has both a return side and a supply side. In the thermal network, the thermal balance equation is composed of thermal generation resources on one side and the thermal load on another side. Equation (9) defines the thermal balance flow.

$$H_{RS,t}^{Boiler} + H_{RS \subseteq RS,t}^{chp-heat} + H_{RL(RS,RS),t}^{thermal-line} = H_{RS,t}^{thermal-load} \quad (9)$$

Thermal flow: There must always be a balance between the production and consumption of thermal energy. To meet thermal demands,

the temperature in the pipelines, mass flow rates, and generation sources in the thermal network are carefully managed. The energy difference between the start of the supply pipes and the end of the return pipelines represents the total thermal energy requirement as indicated in equation (10) in line with the first law of thermodynamics. This is because thermal energy is transferred from the sources to the loads through thermal pipelines, using hot or cold water as the primary thermal transfer medium. Additionally, the temperature at a given node is different from the temperature at the start node of the pipeline, as demonstrated in equation (11). Thermal transmission is dynamic, meaning it has transmission delays that depend on the mass flow rate and the length of the pipeline. This dynamic effect can be illustrated by the vertical section of a supply pipeline.

$$H_{RL,t}^{thermal-line} = G_{RL} (T_{RL}^S - T_{RL}^R) \quad (10)$$

$$T_{RL}^S - T_{RL}^R = \frac{P_{gas-line} C_p \Delta T_{gas}}{C_{fuel}} \quad (11)$$

- Thermal energy convertors

CHP unit: The generation of electricity and thermal in CHP systems is represented using a convex polyhedron model. In a CHP operation, the thermal output can be approximated linearly based on the loading percentage and efficiency. Combined Heat and Power (CHP) systems can simultaneously produce both electrical and thermal energy from a single fuel source. Their thermodynamic performance characteristics allow us to calculate the overall amount of electricity generated. Specifically, the relationship between fuel input, electrical efficiency, and thermal recovery is represented by equations (12)–(14) from Ref. [26], which can be used to accurately determine the total electricity output of the CHP system.

$$P_{e',t}^{chp-heat} = \sum_q \alpha_q P_q^{input} \eta_{chp} \quad (12)$$

$$P_q^{input} = \frac{P_{gas-chp}}{K_1} \quad (13)$$

$$M_{chp} \underline{C}^{chp-heat} \leq P_{chp-heat}, H_{chp-heat} \leq \overline{C}^{chp-heat} M_{chp} \quad (14)$$

The amount of electricity produced from the natural gas input ($P_{gas-chp}$) was calculated using the equation for CHP. The thermal requirement will be converted using the remaining electricity. Equation (15) depends on the input gas being precisely between the previous and subsequent values. To calculate the electrical power output, one should use Equation (10), which defines a specific operating block of the CHP characteristics curve. Additionally, equations (16) and (17) in Ref. [26] illustrate the relationship between fuel consumption, heat recovery, and operating efficiency, enabling the numerical quantification of the equivalent thermal output of the CHP unit.

$$\underline{C}^{gas-chp} \leq P_{gas-chp} \leq \overline{C}^{gas-chp} \quad (15)$$

$$H_{RS \subseteq RS,t}^{chp-heat} = \sum_q \alpha_q H_q^{input} \eta_{chp} \quad (16)$$

$$H_q^{input} = \frac{P_{gas-chp}}{K_2} \quad (17)$$

Boiler unit: The performance of a boiler unit is crucial for achieving optimal energy conversion and system efficiency within an IWPS. To meet the varying heating demands of the system, the boiler requires a consistent supply of gas to convert chemical energy into thermal energy. Enhancing boiler operational efficiency is essential for reliable performance, effective fuel cost management, and minimizing energy losses. The thermal generated by the boiler unit can be calculated by equation

(18) regarding the boiler input gas. The thermal balance flow (equation (9)) examines how boiler operations interact with other components of the system to improve overall energy usage and resilience in IWPS. Equations (19) and (20) also explore the limitations that influence boiler outputs and highlight the set of total gas input.

$$H_{RS,t}^{Boiler} = \eta_{boil} \frac{p_{gas-boiler}}{K_3} \quad (18)$$

$$\bar{C}^{Boiler} \leq H^{Boiler} \leq \bar{C}^{Boiler} \quad (19)$$

$$p_{gas-total} = p_{gas-line} + p_{gas-boiler} + p_{gas-chp} \quad (20)$$

3) Realistic Water Network Modelling

The water network is represented as a directed graph, denoted as $(ww; wl)$, where ww refers to the set of nodes and wl refers to the set of pipes. The nodes are categorized into three disjoint subgroups: tanks (ww') , and junctions (ww, ww') . In the water network, the water flow at each intersection must satisfy both the specified water demand $(W_{ww,t}^{water-load})$ and the continuity condition as shown in (21). Also, $W_{ww,t}^{grid-up}$ simulates the transformer unit's delivery of water from the upstream grid. This is a resource that generates inputs for the water grid.

$$W_{ww,t}^{grid-up} + W_{wl(ww,ww'),t}^{wat-line} + W_{ww \subseteq ww,t}^{desalination} = W_{ww,t}^{water-load} \quad (21)$$

Water flow: The Darcy-Weisbach equation in (22) describes hydraulic heads and flow rates in various pipes, where $F_{wl} > 0$ represents the constant friction factor for pipe wl and $h_{ww,t}$ is hydraulic.

$$h_{ww,t} - h_{ww',t} = F_{wl} \left(W_{wl(ww,ww'),t}^{wat-line} \right)^2 \quad (22)$$

As discussed in Ref. [23], the non-convex quadratic constraint found in equation (23) can be transformed into a second-order cone (SOC) formulation. This transition into a convex optimization framework allows for the identification of effective and manageable solutions. To maintain linearity and ensure tractability, equation (23) employs a convex relaxation of the pressure-flow relationship in the water network. This modeling technique builds upon the work presented in Ref. [23], where pipeline dynamics are approximated to enhance optimization feasibility. Although this relaxation simplifies the original non-convex formulation, validation experiments have shown that the resulting gap is negligible under normal operating conditions. Further research is recommended to evaluate the tightness of the relaxation in more depth.

$$h_{ww,t} - h_{ww',t} = F_{wl} W_{wl(ww,ww'),t}^{linear-line} \quad (23)$$

$$W_{wl(ww,ww'),t}^{linear-line} \geq 0 \quad (24)$$

$$W_{wl(ww,ww'),t}^{linear-line} \leq W_{wl(ww,ww'),t}^{wat-line} W_{wl}^{base} \quad (25)$$

$$W_{wl(ww,ww'),t}^{linear-line} \geq 2 W_{wl}^{base} \quad (26)$$

Where $W_{wl(ww,ww'),t}^{linear-line}$ is auxiliary variables and W_{wl}^{base} is the limitation of pipe wl .

- Water energy convertors

Desalination unit: Given the varying hydraulic heads at a desalination tank's inlet and outlet, we define the set of tank nodes into two subsets: (ww) for the inlet nodes and (ww') for the outlet nodes, maintaining the same cardinality. There is no conduit connecting node (ww) to node (ww') . The outlet head of the tank changes over time as:

$$h_{ww',t} = h_{ww,t-1} + \frac{\Delta t}{S} \left(W_{ww' \subseteq ww,t}^{sea} - W_{ww' \subseteq ww,t}^{desalination} \right) \quad (27)$$

Let (Δt) represent the cross-sectional area of the tank, and let (t) denote the length of each time slot. The outlet head varies with time according to the established parameters. Additionally, we have a constant for the initial head, $(h_{ww,t}^0)$. The output head is subject to limitations: it cannot exceed the inlet head and is constrained from below by a defined constant (S) . Power $(P_{e' \subseteq e,t}^{desalination})$ is utilized by pumps to raise the hydraulic head along pipes.

$$h_{ww',t} - h_{ww,t} \leq -A_{wl} \left(W_{ww' \subseteq ww,t}^{desalination} \right) + B_{wl} \left(W_{ww' \subseteq ww,t}^{desalination} \right) + C_{wl} \quad (28)$$

$$P_{e' \subseteq e,t}^{desalination} = \frac{p g}{\eta_{des}} W_{ww' \subseteq ww,t}^{desalination} \quad (29)$$

$$0 \leq W^{desalination} \leq \bar{C}^{desalination} M_{desalination} \quad (30)$$

In this context, the parameters p, g , along with the constants A_{wl}, B_{wl} , and C_{wl} , are all positive. The power consumption of the pump at time t , represented as $P_{e' \subseteq e,t}^{desalination}$, depends on the gravitational coefficient g , the water density p , and the constant pump efficiency η_{des} . Thus, equations (28)–(30) establish the connection between the water and power networks.

3. Attack model for Iwps

The attack modeling for IWPSs is discussed in this section. State estimation is an effective technique for identifying FDIA by analyzing raw data measurements; however, an adversary's covert design may render a successful FDIA invisible. Modeling attacker activity using time-series data could provide valuable insights into the long-term vulnerabilities of systems. However, this study focuses on False Data Injection Attacks (FDIAs) as static disturbances to evaluate the structural risk of the Integrated Water Distribution System (IWPS). By using a static formulation, we can effectively analyze large-scale, multi-layered systems, making it computationally feasible to conduct a detailed examination of system sensitivity and optimization under specific attack scenarios. Future research will expand on this approach to include dynamic FDIA scenarios with temporal propagation, utilizing stochastic or predictive attacker models.

1) Attack on the electricity network

The nonlinear relationship between the measurement, z , and the state variable, X , is described in equation (31), where e , represents the measurement error and $h(X)$ is the nonlinear vector function of X . This relationship can be transformed into equation (32) through DC state estimation, with H denoting the Jacobian matrix.

$$z = h(X) + e \quad (31)$$

$$z = HX + e \quad (32)$$

The estimated state vector can be expressed as $\hat{X}_{bad} = X_{bad} + c$, where c denotes the resultant deviation vector of the state variable following FDIA. After the FDIA is implemented, the measurement vector z , is modified to $z_{bad} = z + a$ where, a , represents the attack vector. Consequently, the estimated state variable can be determined by formulating, $\hat{X}_{bad}$, as follows:

$$z_{bad} = h(X_{bad}) + e \quad (33)$$

Load measurement can be viewed as a specific example of FDIA by manipulating the data attack vector sum to equal zero, as referenced in equations (34) and (35). For clarity, we use the term Δ to describe the FDIA affecting electrical data. To maintain the attack within detectable

or acceptable levels, the Attack Injection Level (AIL), specified in equation (36), imposes a maximum limit on the magnitude of the altered signal, thereby quantitatively constraining the deviation caused by any cyberattack.

$$\sum \Delta \left(p_{e,t}^{transfor-up}, p_{e,t}^{elec-line}, p_{e',t}^{chp-heat}, p_{e,t}^{battery}, p_{e,t}^{elec-load}, p_{e',t}^{desalination} \right) = 0 \quad (34)$$

$$\sum \Delta (V_e, \delta_e) = 0 \quad (35)$$

$$\underline{AIL} \leq \left(p_{e,t}^{transfor-up}, p_{e,t}^{elec-line}, p_{e',t}^{chp-heat}, p_{e,t}^{battery}, p_{e,t}^{elec-load}, p_{e',t}^{desalination} \right) \leq \overline{AIL} \quad (36)$$

2) Attack on the thermal network

In a thermal system, FDIA is relevant for both density measurement and gas load. The FDIA for thermal load measurement and density is detailed in equations (37) and (38). These are comparable to the FDIA applied to electricity data measurement. The attack coefficient, denoted as β_g , changes due to the modified gas density measurement. Consequently, under FDIA conditions, the initial coefficient in equation (39) is altered to the form presented in equation (40). Attack Injection Level (AIL) limitations are applied to both electrical and thermal networks to accurately simulate and identify Forged Data Injection Attacks (FDIA) within the Integrated Water and Power System (IWPS). By defining the upper and lower limits of attack magnitude, these constraints ensure that the model focuses on subtle yet effective attacks. Additionally, equations (34), (35) and (37) enforce the requirement that the sum of the changes ($\sum \Delta$) equals zero. This maintains the model's physical plausibility and helps differentiate malicious redistributions from genuine system defects. This approach allows for the modeling of sophisticated cyberattacks that traditional threshold-based detectors might miss by ensuring that the total injected variation does not alter the net power or energy balance.

$$\sum \Delta \left(H_{RS,t}^{Boiler}, H_{RS \subseteq RS,t}^{chp-heat}, H_{RL(RS,RS1),t}^{thermal-line}, H_{RS,t}^{thermal-load} \right) = 0 \quad (37)$$

$$\sum \Delta (p_{gas-total}) = 0 \quad (38)$$

$$\underline{AIL} \leq \left(H_{RS,t}^{Boiler}, H_{RS \subseteq RS,t}^{chp-heat}, H_{RL(RS,RS1),t}^{thermal-line}, H_{RS,t}^{thermal-load} \right) \leq \overline{AIL} \quad (39)$$

$$0 \leq (p_{gas-total}) \leq (1 + \beta_g) p_{gas-total} \quad (40)$$

3) Attack on the water network

FDIA in a water system is important for dependency and independence variables defined as $W_{ww,t}^{grid-up}$, $W_{wl(ww,ww1),t}^{wat-line}$, $W_{ww' \subseteq ww,t}^{desalination}$. Indeed, an attacker can change these variables through the change in the state variable $h_{ww,t}$ or the input parameters $W_{ww,t}^{water-load}$. Equations (41) and (42) describe the FDIA for the water network. When it comes to measuring power networks, these are similar to the FDIA. Because of the altered water density measurement ($h_{ww,t}$), the attack coefficient, represented by the symbol β_w , varies. As a result, the initial coefficient in equation (43) takes on the shape seen in equation (44) under FDIA conditions.

$$\sum \Delta \left(W_{ww,t}^{grid-up}, W_{wl(ww,ww1),t}^{wat-line}, W_{ww' \subseteq ww,t}^{desalination}, W_{ww,t}^{water-load} \right) = 0 \quad (41)$$

$$\sum \Delta (h_{ww,t}) = 0 \quad (42)$$

$$\underline{AIL} \leq \left(W_{ww,t}^{grid-up}, W_{wl(ww,ww1),t}^{wat-line}, W_{ww' \subseteq ww,t}^{desalination}, W_{ww,t}^{water-load} \right) \leq \overline{AIL} \quad (43)$$

$$0 \leq (h_{ww,t}) \leq (1 + \beta_w) h_{ww,t} \quad (44)$$

4. Risk assessment model

To ensure the reliability and security of Integrated Water, Power, and Thermal Systems (IWPS), it is crucial to conduct a risk assessment that identifies potential threats and system vulnerabilities. These interconnected networks are sensitive to fluctuations in demand, component failures, and external factors such as cyberattacks, which can lead to cascading effects across the entire system. The Jacobian matrix plays a key role in understanding the interdependencies between the networks, as it quantifies the sensitivity of one network to disturbances in another. By analyzing the condition number and eigenvalues of the Jacobian matrix, one can assess the system's sensitivity and stability thresholds. A high condition number indicates high sensitivity to disturbances, while eigenvalue analysis reveals potential instability. These insights are crucial for designing risk-reduction strategies, such as adaptive control systems and redundancy in critical components. To effectively apply this approach, it is necessary to define the state and control variables for each network. The state variables represent the system's operational state, while control variables refer to the adjustable parameters that influence the system's behavior. Through Jacobian matrix analysis, risk management can be enhanced, ensuring the resilience and continued reliable operation of IWPS under various conditions.

$$X = [\chi_{elec} \chi_{therm} \chi_{wat}]$$

$$Y = [Y_{elec} Y_{therm} Y_{wat}]$$

For electricity network:

$$\chi_{elec} = \{V_e, \delta_e\}$$

$$Y_{elec} = \{p_{transfor-up}, p_{elec-line}, p_{chp-heat}, p_{battery}, p_{desalination}\}$$

For thermal network:

$$\chi_{therm} = \{p_{gas-total}, T_{RL}^S\}$$

$$Y_{therm} = \{H_{RS,t}^{Boiler}, H_{RS \subseteq RS,t}^{chp-heat}, H_{RL(RS,RS1),t}^{thermal-line}\}$$

For water network:

$$\chi_{wat} = \{h_{ww,t}\}$$

$$Y_{wat} = \{W_{ww,t}^{grid-up}, W_{wl(ww,ww1),t}^{wat-line}, W_{ww' \subseteq ww,t}^{desalination}, W_{ww,t}^{sea}\}$$

The Jacobian matrix for a multi-state variable system is presented in its normal form in equation (45). Each entry of the Jacobian matrix, including both diagonal and off-diagonal elements, is computed to capture the system's output sensitivity relative to its inputs and internal states. This detailed approach highlights how changes to one variable impact the system as a whole. The structural form of the Jacobian matrix for the Integrated Water and Power System (IWPS) is formally defined by equations (46)–(48), which represent the interdependencies and cross-couplings between control signals, electrical power variables, and water flow rates.

$$JM = \begin{bmatrix} \left(\frac{\partial Y_{sel}}{\partial \chi_{sel}} \right) & \dots & \left(\frac{\partial Y_{ns}}{\partial \chi_{sel}} \right) \\ \vdots & \ddots & \vdots \\ \left(\frac{\partial Y_{sel}}{\partial \chi_{ns}} \right) & \dots & \left(\frac{\partial Y_{ns}}{\partial \chi_{ns}} \right) \end{bmatrix} \quad (45)$$

$$\frac{\partial Y_{sel}}{\partial \chi_{sel}} = \left(\frac{\partial f_{elec}}{\partial Y_{sel}} \right)^{-1} \left(\frac{\partial Y_{sel}}{\partial \chi_{sel}} \right) \quad (46)$$

$$\frac{\partial Y_{sel}}{\partial \chi_{ns}} = \left(\frac{\partial Y_{sel}}{\partial Y_{sel}} \right) \left(\frac{\partial \chi_{sel}}{\partial Y_{ns}} \right) \left(\frac{\partial Y_{ns}}{\partial \chi_{ns}} \right) \quad (47)$$

For this case:

$$JM = \begin{bmatrix} \frac{\partial Y_{elec}}{\partial \chi_{elec}} & \frac{Y_{therm}}{\partial \chi_{elec}} & \frac{Y_{wat}}{\partial \chi_{elec}} \\ \frac{\partial Y_{elec}}{\partial \chi_{therm}} & \frac{Y_{therm}}{\partial \chi_{therm}} & \frac{Y_{wat}}{\partial \chi_{therm}} \\ \frac{\partial Y_{elec}}{\partial \chi_{wat}} & \frac{Y_{therm}}{\partial \chi_{wat}} & \frac{Y_{wat}}{\partial \chi_{wat}} \end{bmatrix} \quad (48)$$

5. the coordinated probability-aware risk model for IWPS

The significance of risk assessment in IWPS cannot be overstated, particularly given their reliance on network connectivity. There is a considerable risk of cascading failures in such systems due to the interaction of numerous variables and subsystems. Various factors can lead to these failures, including operational mistakes, environmental conditions, and, most notably, cyberattacks. Therefore, it is essential to understand and evaluate these hazards to ensure operational continuity, safety, and reliability. To enhance the assessment of system vulnerabilities, risk assessment can be effectively combined with mathematical techniques such as the Jacobian matrix. To understand how cyberattacks can exploit vulnerabilities in one network to disrupt another, it is crucial to recognize the dependencies between them. To mitigate the risks of cyberattacks in IWPS, a risk assessment approach based on the Jacobian matrix can be particularly effective. Fig. 2 illustrates a

flowchart of the proposed framework for IWPS.

1) formulation definition

In the case of the risk model, deriving a final coefficient of the Jacobian matrix to represent the overall performance of the network in response to unforeseen changes is essential for effective management and risk assessment. This process is facilitated through the Jacobian matrix, which encapsulates the relationships between various state and control variables. The mathematical relationship, represented as equation (49), captures how variations in the elements of the Jacobian correspond to unexpected inputs, providing a framework to analyze the overall system response.

$$\mathcal{S}_\beta = \sqrt{\sum_i \sum_j (JM_{ij})^2} \quad (49)$$

Where JM_{ij} is the value of element ij in the matrix JM and the Jacobian coefficient is modeled as $\mathcal{S}_\beta$. Equation (49) referenced likely formalizes the calculation of an overall representative coefficient that captures the behavior of a system in response to changes in state variables, utilizing the Jacobian matrix JM . Computing the matrix norm ($\|JM\|$) yields a scalar quantity that quantifies the system's overall sensitivity to disturbances, demonstrating how changes in state variables propagate to control variables. A high Jacobian norm indicates system vulnerabilities where minor input variations may disproportionately impact performance, proving particularly valuable for identifying potential instabilities. Combining this approach with probabilistic risk assessment

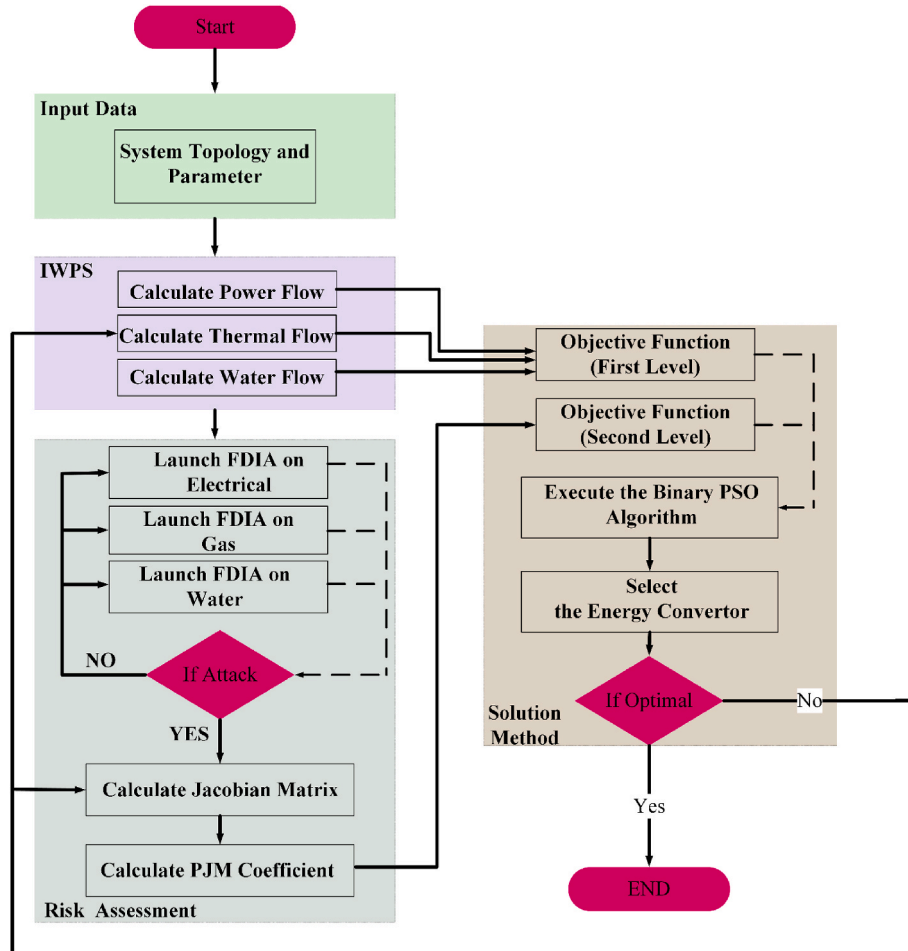


Fig. 2. Illustration of proposed framework flowchart.

creates a comprehensive framework for system resilience management. Within this framework, threats such as cyberattacks or sudden load changes are analyzed using the Jacobian matrix and High-probability low impact events and Low-probability high impact events are prioritized. This integrated methodology enables, identification of critical system weak points, optimal allocation of control resources and development of adaptive strategies for disturbance mitigation. While not capturing every system complexity, this approach provides an effective means for analyzing dynamic behavior and enhancing stability in complex power systems.

The Kalman filter provides a probabilistic framework for updating state estimates using observable data and system models, employing covariance matrices to represent uncertainty [27]. Equations (50) and (51) show the probability function inspired by the Kalman filter approach with state estimation X_k and covariance matrix σ_k^2 .

$$X_k \sim N(X_k^{mesu}, \sigma_k^2 I_k) \quad (50)$$

$$\eta_k = (X_k - X_k^{mesu})^T (X_k - X_k^{mesu}) \quad (51)$$

The recursive estimation framework of the Kalman filter can be utilized to estimate the likelihood of various cyberattack events. The Kalman filter calculates the covariance matrix σ_k^2 , which indicates the uncertainty in state estimations X_k , by predicting the current state of IWPS based on the measurement data X_k^{mesu} and then updating this state with new observations. This approach can model the impact of cyberattacks on different state variables and can linearize system changes when combined with the Jacobian matrix JM . Meanwhile, the probability of each cyber attack κ is calculated as η_k . This combined analysis enables improved risk management and resilience planning by identifying control variables that are most affected by changes due to an attack. All in all, to effectively evaluate risk in IWPSs, it is essential to compute a representative coefficient that reflects the system's sensitivity to changes in state. This can be achieved using the Jacobian matrix, which measures how various control variables respond to shocks. By combining this coefficient with the likelihood of specific cyberattacks, we can develop a more comprehensive understanding of risk.

$$risk = PJM = \sum_k \eta_k \mathcal{J}_\beta^\kappa \quad (52)$$

One useful technique for this analysis is defined by equation (52), in which κ is the number of cyber-attacks launched in IWPS. By integrating this approach, we can establish a robust framework for risk assessment, ultimately enhancing system resilience and enabling proactive defense planning.

2) Energy Converters-Based Robust Energy Strategies

To achieve a robust energy framework, it is essential to reduce the risk of cyberattacks. The Jacobian matrix can be utilized to model how the IWPS responds to changes in state variables during a comprehensive risk assessment. By evaluating the likelihood of various cyberattack scenarios, operators can identify and protect vulnerable points within the network and assess the potential impact of these attacks on critical network components. To effectively mitigate risk, it is crucial to implement defensive measures, enhance critical infrastructure, and ensure that the system can dynamically adapt to maintain stability and reliability in the face of potential cyber threats. A robust energy management plan that incorporates energy converters and advanced risk analysis can enhance response times, increase system stability, and optimize energy distribution across multiple networks. This holistic approach is essential for advancing the resilience and sustainability of modern systems. For IWPS that incorporate various energy carriers and converters, such as gas, water, and electricity, effective energy management is crucial for maximizing reliability and performance. To facilitate efficient transfers between different energy carriers, IWPS

utilizes a range of energy converters, including boilers, CHP units, and electric heaters. Therefore, it is important to develop a thorough plan that considers both the operational efficiency of each individual unit and the overall resilience of the system against operational risks and disruptions. Implementing hybrid energy strategies that integrate three additional energy systems is essential for effectively managing the operations of water, heating, and electrical grids within IWPSs. The proposed systems, 1) Photovoltaic (PV) unit, 2) water-to-thermal conversion (WH), 3) electricity-to-thermal conversion (EH), and 4) thermal storage (HS) aim to address the significant vulnerability of IWPSs to fluctuations in the thermal grid.

- PV unit

The PV system is a vital element of the DC microgrid. Numerous studies have demonstrated a clear relationship between the power generated by the PV system and the level of solar radiation, as shown in equation (53). Increased solar radiation correlates with higher PV output power, up to the system's maximum capacity. Equation (53) takes these factors into account and includes a term for PV system losses to ensure precise power generation calculations.

$$PV = \frac{E C^{PV} M_{PV}}{g} (1 - loss^{PV}) \quad (53)$$

- EH unit

For energy systems, particularly IWPSs, the conversion process is essential because it allows for the efficient use of electricity for heating. This efficiency helps balance energy supply and demand across various carriers. Additionally, reducing the reliance on fossil fuels for heating is a significant factor in developing a more sustainable energy system. Using the input power ($P^{electricity}$), equations (54) and (55) illustrate the output thermal (HP^{heat}) from the IWPS [33].

$$HP^{heat} = \mu \eta_H P^{electricity} \quad (54)$$

$$0 \leq P^{electricity} \leq \bar{C}^{electricity} M_{EH} \quad (55)$$

- WH unit

WH conversion is an environmentally friendly process that plays a crucial role in integrated energy systems, particularly in IWPSs that prioritize efficient and renewable energy use. This procedure involves using thermal exchangers or geothermal pumps to transfer thermal energy from a thermal source to various applications, including space heating, industrial processes, and domestic hot water systems. In this context, the heat generated by the WH unit is denoted as (Hw^{heat}), while the water input is represented by the variable (W^{water}). The WH unit converts water energy into heat energy based on its efficiency (η_W) and a constant (β). The WH unit is described by equations (56) and (57). A crucial element of IWPSs designed for efficient and renewable energy consumption is the WH converter. This sustainable process takes advantage of water's high heat capacity to effectively store and transport energy.

$$Hw^{heat} = \beta \eta_W W^{water} \quad (56)$$

$$0 \leq W^{water} \leq \bar{C}^{water} M_{WH} \quad (57)$$

- HS unit

The process of absorbing and retaining thermal energy for future use is known as HS. This is typically accomplished using thermal-retaining materials or systems, such as phase-change materials, water tanks, or

specialized thermal batteries. Thermal storage is particularly important in IWPS that rely on intermittent renewable energy sources like solar or wind power. By storing excess thermal during periods of low demand and releasing it when needed, thermal storage improves system reliability, enhances energy efficiency, and allows for load shifting. These aspects are all critical for sustainable energy management. The level of thermal energy in HS must be limited based on the charge and discharge thermal values (H_t^{chr}, H_t^{dichr}), as shown in equation (58). The limitations of the HS unit are modeled by (59)–(61).

$$E_{t-}^{H-storage} = E_{t-1}^{H-storage} + H_t^{chr} - H_t^{dichr} \quad (58)$$

$$H_t^{H-storage} = H_t^{dichr} - H_t^{chr} \quad (59)$$

$$\underline{H}_t^{dichr}, \underline{H}_t^{chr} \leq H_t^{dichr}, H_t^{chr} \leq \bar{H}_t^{dichr}, \bar{H}_t^{chr} \quad (60)$$

$$\underline{E}_t^{H-storage} \leq E_t^{H-storage} \leq \bar{E}_t^{H-storage} \quad (61)$$

It is essential to evaluate different combinations of energy sources and converters, such as desalination, CHP, WH, EH, PV, and SH, to develop a safe and cost-effective IWPS. Each combination of these components presents a distinct approach that may help balance operational costs and cybersecurity risks while optimizing overall system performance. When examining these tactics, two primary goals should be considered. First, it is essential to identify a combination of strategies that can preserve or enhance the system's defenses against potential cyberattacks. Technical and Operational Power Systems IWPSs are vulnerable to cyber threats that could lead to significant disruptions or even halt operations, particularly due to their increasing reliance on digital control and communication infrastructures. Therefore, evaluating the resilience of an energy system requires an assessment of the potential impacts of cyberattacks on the functioning of each energy source, converter, and interconnections. Reducing energy expenses remains a top priority for operations. This involves identifying which desalination, CHP, WH, EH, PV, and SH provide the best financial savings without compromising the overall robustness and reliability of the system. Each energy source has its own unique benefits and limitations. For example, while CHP systems may have higher upfront costs, they are well-known for their efficiency in generating both thermal and electricity. PV systems, although dependent on weather conditions, offer renewable energy with low operating costs. EH and SH provide greater flexibility to meet fluctuating demand, while WH systems can deliver effective energy recovery solutions. To conduct an effective analysis, it is essential to model various configurations of these five energy sources under different circumstances. This involves creating models that take into account variables such as system integration, energy output, and responses to simulated cyberattacks. The goal is to identify trends where specific combinations consistently offer better protection and lower costs. For example, desalination, CHP, PV units, and WH may result in a well-balanced system that enhances resilience by utilizing distributed generation and diverse energy sources. Additionally, incorporating SH and EH can provide further redundancy and options for load management. In this study, sensitivity analysis parameters (PJM) are utilized to present a combination of energy resources and converters (Π) and to outline various techniques for responding to cyberattacks on IWPSs. The operator of an IWPS must choose one of these strategies to manage energy as efficiently as possible during cyberattacks. The complete set of resilient strategies (Ω) available for IWPSs is detailed in equation (62) and (63), with an emphasis on maintaining energy balance within the relevant IWPS.

$$\Pi = \{M_{chp}, M_{desalination}, M_{WH}, M_{EH}, M_{PV}\} \quad (62)$$

$$\Gamma = \{\Pi_1, \Pi_2, \dots, \Pi_n\} \quad (63)$$

3) A ROBUST PROBABILITY-AWARE RISK MANAGEMENT

To determine the optimal operation of IWPS, we have developed a two-level mathematical framework with two primary objectives: reducing operating expenses and minimizing the system's risk coefficient. In this framework, each energy convertor such as CHP, WH, EH, PV, and desalination is represented by a binary variable. If this variable is set to 1, it means the energy source is included in the system's operation; if it is set to 0, it means it is not included. The balancing equations for interconnected networks, which involve thermal, water, and electricity, are adjusted based on the addition or removal of these energy sources. This allows each network to adapt dynamically to the system design, depending on the selected combination of sources. Hence, equations (64)–(67) indicate the first level of the problem along with the objective function and the corresponding constraints.

First level:

$$\min \sum_k \eta_k \left(\sqrt{\sum_i \sum_j (JM_{ij})^2} \right)_k \quad (64)$$

s.t.

$$\Gamma = \{\Pi_1, \Pi_2, \dots, \Pi_n\} \quad (65)$$

$$\Pi = \{M_{chp}, M_{desalination}, M_{WH}, M_{EH}, M_{PV}\} \quad (66)$$

$$\sum_{\zeta=1}^n \Gamma_{\zeta} = 1 \quad (67)$$

This two-level approach begins with a focus on risk minimization, specifically aimed at reducing the IWPS's vulnerability to potential disruptions and cyber threats. By assessing the vulnerabilities across various system components and considering the likelihood of cyberattacks, we can assign a risk coefficient that reflects the network's resilience. To protect the IWPS from attacks that could lead to costly outages or even cascading failures across interconnected networks, the framework aims to lower this risk coefficient. In this context, Π represents a set of energy converters that could be utilized within the system. Meanwhile, Γ denotes a collection of possible combinations of these energy converters. A specific status in Γ can be identified as the robust approach based on the criteria outlined in equation (64).

Second level:

$$\min \sum (C_g P^{transfor-up} + C_g PV) + (C_{gas} P^{gas-total}) + (C_w W^{grid-up}) \quad (68)$$

s.t.

$$\begin{aligned} P_{e,t}^{transfor-up} + P_{e,t}^{elec-line} + P_{e,t}^{chp-heat} M_{chp} + P_{e,t}^{battery} + P_{e,t} M_{PV} \\ = P_{e,t}^{elec-load} + P_{e,t}^{desalination} + P_{e,t}^{electricity} M_{EH} \end{aligned} \quad (69)$$

$$\begin{aligned} H_{RS,t}^{Boiler} + H_{RS,t}^{chp-heat} + H_{RS,t}^{thermal-line} + H_{RS,t}^{heat} M_{EH} + H_{RS,t}^{heat} M_{WH} + H_{RS,t}^{H-storage} \\ = H_{RS,t}^{thermal-load} \end{aligned} \quad (70)$$

$$W_{ww,t}^{grid-up} + W_{ww,t}^{wat-line} + W_{ww,t}^{desalination} M_{desalination} = W_{ww,t}^{water-load} + W_{ww,t}^{water} M_{WH} \quad (71)$$

$$\text{Eqs. (2)–(8), (10)–(20), (22)–(30)} \quad (72)$$

- $C_g P^{transfor-up}$: electricity purchased from the upstream electrical grid,
- $C_g PV$: operational and maintenance cost of the photovoltaic system,
- $C_{gas} P^{gas-total}$: natural gas consumption for CHP and boiler operation,
- $C_w W^{grid-up}$: water procurement cost from upstream sources or desalination units.

The second level of the framework focuses on reducing operating costs. It encompasses all financial aspects of energy sources and their conversions, including fuel prices, maintenance costs, and other

operating expenditures. By minimizing these costs, the framework identifies a cost-effective operational strategy that meets demand across all networks. The goal function for this level is defined in equation (68), which specifies the input energy prices for the three networks: water, thermal, and electricity. The new energy converters, whether functioning as generation or load, help to redefine the energy balance for each network. Equations (69)–(72) illustrate the energy balances for water, thermal, and electricity, respectively.

All in all, this two-level optimization methodology serves as a strategic tool for evaluating IWPS configurations. It aids decision-makers in selecting the optimal operational approach that balances increased resilience against cyberattacks with cost-effectiveness. In complex energy systems, where decisions must factor in both financial constraints and security threats, this comprehensive approach proves highly beneficial. The model's flexibility makes it well-suited for dynamic, networked systems, where configurations often require adjustments to accommodate changing energy demands, evolving cyber threats, and budgetary limitations.

4) solution method

It is recognized that there can be conflicts between risk avoidance and reducing operational expenses in certain scenarios, such as peak demand, limited resource availability, or heightened cyber threat levels. The proposed architecture includes an adjustable risk-cost coefficient along with an adaptive multi-objective optimization approach to address this issue. This allows for real-time modifications of optimization priorities based on current system conditions. Additionally, a set of optimal solutions is generated using a Pareto-based approach, enabling system operators to select the best cost-risk ratio in line with their operational regulations. The proposed mathematical risk management framework effectively utilizes the binary Particle Swarm Optimization (PSO) method to identify the optimal energy model [28]. In this approach, each potential configuration is represented as a particle in a solution space, where the binary values of each particle indicate the presence or absence of specific energy converters. This binary PSO method enables an efficient search for the best combinations Γ . The risk management framework aims to reduce the risk coefficient and operating expenses, and it successfully achieves these objectives. By iteratively adjusting the positioning of particles based on both individual and global best solutions, the method identifies configurations that not only satisfy energy demands but also enhance IWPS resilience against cyberattacks while remaining cost-effective. In the case of PSO, each particle is represented as a point in D-dimensional space, where the i th particle corresponds to a D-dimensional vector ($X^i = (X^1, X^2, \dots, X^D)$). In this context, the search space is D-dimensional and consists of a swarm of i particles, where X^i is defined as the set of energy converters, M_{chp} , $M_{desalination}$, M_{WH} , M_{EH} , M_{PV} . This setup suggests that each particle could represent a potential global optimum of the function ($f(x)$) within the specified domain D. The evaluation of each particle is done using ($f(x)$), with the particle's positional coordinates serving as the input values. The output value is often called the fitness value. The associated operators update each particle's position, guiding them toward better regions until the optimal point is reached based on this fitness value. The best location found by the entire swarm is recorded as p_g^k , while each particle's previous best position is remembered and represented as p^k throughout the iterative process. The "flying" velocity of the i particle is represented as a D-dimensional vector, denoted as v_k^i . At each step, the velocities of all particles are updated by multiplying their current velocities, local best values, and global best values by three constants, w , c_1 , c_2 respectively, as shown in equations (73)–(76). Moreover, each particle's position is updated by adding its velocity to its current position, as indicated in Eq. (2).

$$v_{k+1}^i = w v_k^i + c_1 (p^k - X_k^i) + c_2 (p_g^k - X_k^i) \quad (73)$$

$$X_{k+1}^i = X_k^i + v_k^i \quad (74)$$

The standard PSO method is not suitable for optimizing purely discrete binary combinatorial problems, as it operates in continuous and real number spaces. To tackle this issue, Kennedy and Eberhart developed the KBPSO algorithm. In this approach, particles are represented as binary vectors of length u , and the velocity determines the probability that a bit X^i will take the value of 1. While the velocity is confined to the interval $[0.0, 1.0]$ through a limiting transformation function, this means that in KBPSO, a particle changes its bit value based on specific criteria. The updating formula for the velocity is preserved (refer to (73)).

$$S(v_k^i) = 1 / (1 + e^{-v_k^i}) \quad (75)$$

$$X_k^i = \begin{cases} 1 & \text{if } rand() \leq S(v_k^i) \\ 0 & \text{otherwise} \end{cases} \quad (76)$$

where the function $S(v_k^i)$ is a sigmoid limiting transformation and the result of $rand$ is taken from $\{0,1\}$.

6. Case study

This section conducts simulations on an Integrated Water and Power System (IWPS) connected to three networks: electricity, thermal, and water. The aim is to verify the effectiveness of the proposed risk models and their impact on the sensitivity function of the IWPSs. We compare the outcomes of the risk model introduced in this paper during IWPS operations with those of conventional risk models found in the literature [19]. Initially, we examine an IWPS that includes 36 buses and 36 lines for the electricity network, along with 5 buses and 5 lines for the thermal and water networks, respectively. Furthermore, the IWPS accounts for variations in water, thermal, and electrical loads over a 24-h period. Specific details can be found in Ref. [21]. The test system used in this study was adapted from Ref. [21] to emulate the structural and functional characteristics of a real Integrated Water and Power System (IWPS). Additionally, the configuration features energy flow channels, interdependent control loops, and a realistic arrangement of converters. While the system is moderately scaled, it offers sufficient complexity to effectively evaluate the proposed cyber-risk analysis and optimization approach in practical scenarios. The operating capacities of the energy converters are sourced from Ref. [10], while additional details about the Integrated Water Power System (IWPS) can be found in Ref. [21]. Fig. 3 illustrates the fundamental design of the system, which was implemented using GAMS and MATLAB on an i7 PC with 64 GB of RAM.

1) The analysis of Jacobian matrix for IWPS

The sensitivity matrix research indicates that the electricity network demonstrates a higher sensitivity level compared to the thermal and water networks. Specifically, the sensitivity values for the electrical network, particularly concerning transformer output ($P^{transfor-UP}$), reveal that even minor adjustments to transformer power settings can't significantly impact the system's voltage (V_e) and angle (δ_e). In Table 2, the effect of line flow on the network's voltage is notably greater than other influencing factors, with a sensitivity value of 161.6068. This suggests that the electrical network is particularly vulnerable to changes in this area. The values of 2.846615 and 2.795893 indicate that other decision variables related to the electrical network, such as CHP power ($P^{chp-heat}$) and desalination power ($P^{desalination}$), despite their smaller magnitudes, still show noticeable sensitivity to voltage fluctuations. This underscores the importance of power-related decisions in influencing the overall behavior of the system. In contrast, the thermal and water networks demonstrate somewhat lower sensitivity, particularly regarding their effects on state variables like water flow (h_{wvt}) and total

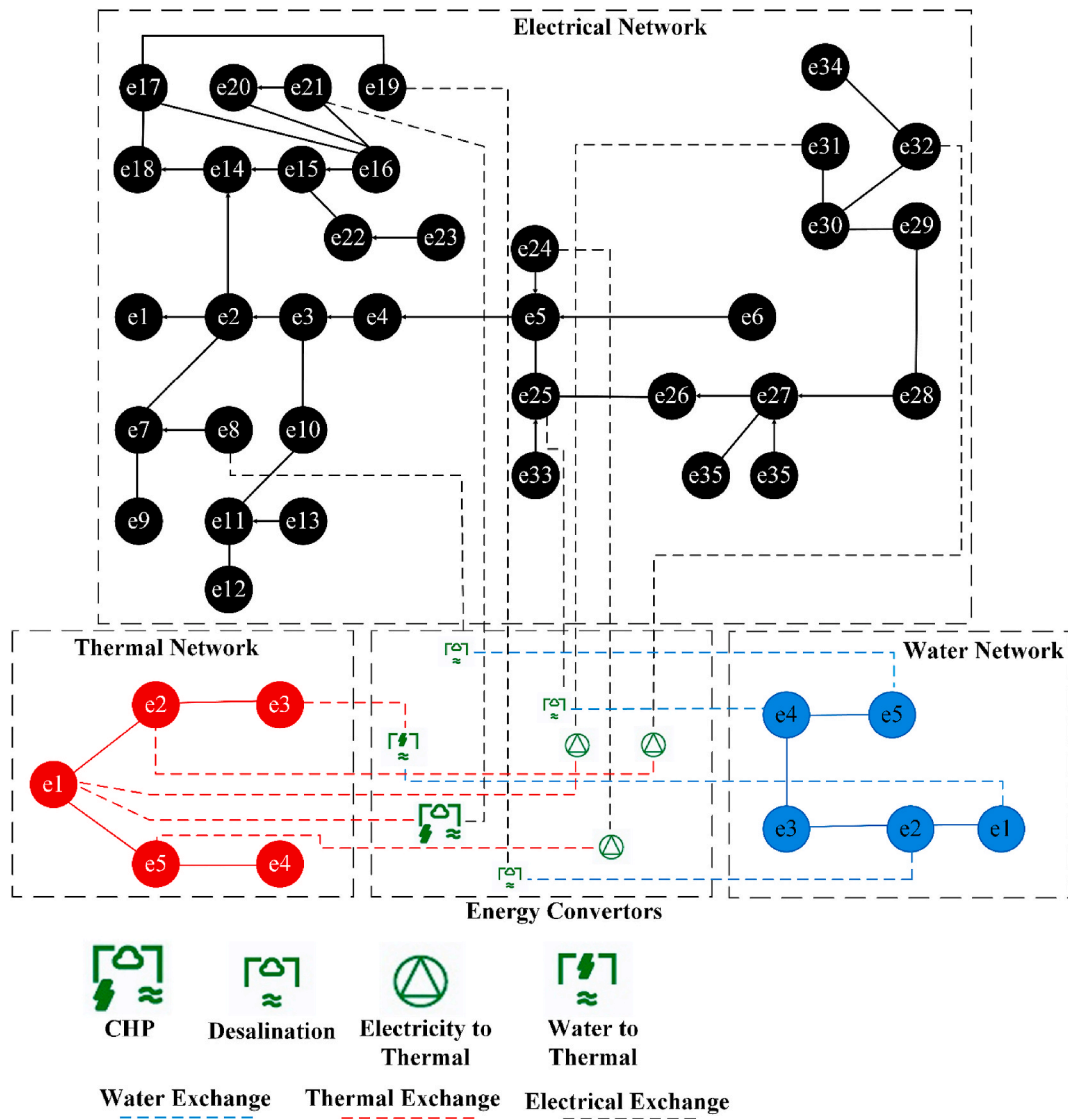


Fig. 3. The fundamental design of the system [29].

Table 2
The output of Jacobean matrix for IWPS.

dY/dx	Electricity Network		Thermal Network		Water Network	Avr.
	V_e	δ_e	$p_{gas-total}$	T_{RL}^s	$h_{ww,t}$	
$p_{transfor-UP}$	0	0	0	0	0	0
$p_{elec-line}$	161.6068	15.8082	209.1542	6.763112	30.53426	84.77331
$p_{chp-heat}$	2.846615	0.278453	3.684136	0.119129	0.537844	1.493235
$p_{battery}$	0.046821	0.00458	0.060596	0.001959	0.008846	0.02456
$p_{desalination}$	2.795893	0.273491	3.61849	0.117006	0.528261	1.466628
Total	17.5515466					
$H_{gBoiler}$	5.423068	0.530478	7.018623	0.226951	1.024644	2.844753
$H_{chp-heat}$	3.912431	0.38271	5.063532	0.163732	0.739221	2.052325
$H_{thermal-line}$	2.458476	0.240485	3.1818	0.102885	0.464509	1.289631
Total	2.062236333					
$w_{grid-up}$	0	0	0	0	0	0
$w_{wat-line}$	0.033582	0.003285	0.043462	0.001405	0.006345	0.017616
$w_{desalination}$	0.201351	0.019696	0.260592	0.008426	0.038044	0.105622
w_{ww}^{sea}	0.033582	0.003285	0.043462	0.001405	0.006345	0.017616
Total	0.0352135					

gas pressure ($p_{\text{gas-total}}$). The output values for boiler thermal ($H_{\text{Boiler}}^{\text{g}}$) and CHP thermal ($H_{\text{chp-heat}}^{\text{h}}$), represented by 5.423068 and 3.912431 respectively, exhibit moderate sensitivity, indicating their significance in impacting gas pressure. The figures for the electrical grid are still significantly higher, especially regarding line flow output. In contrast, the water network exhibits the lowest sensitivity values, as indicated by values such as 0.210153 and 0.033582. These values suggest that factors related to desalination and water flow have minimal impact on changes in water flow. Since the electricity network is the most responsive component of the system, it is crucial to focus on optimizing decisions related to electricity generation, transmission, and storage. This approach will help reduce risks and improve overall system stability, as evidenced by the significant differences in sensitivity levels. The final column, “Avr.,” indicates the row-wise average sensitivity, which reflects the total impact of each choice variable across all three networks. A high mean value in this column suggests that the associated control variable significantly affects multiple subsystems, making it an important area for system monitoring or potential attack entry. The “Total” rows at the bottom of each subsystem section also highlight the aggregated vulnerability level of each network, providing a summary of the column-wise average sensitivities. These totals offer a comparative perspective on which subsystem is more susceptible to cyber threats, such as False Data Injection Attacks (FDIA), due to its greater sensitivity to changes in decision variables. The overall sensitivity value of the power network is 17.5515466. Among the three networks, the electricity network is the most sensitive, as indicated by its highest sensitivity value. This elevated total sensitivity demonstrates that the state variables, particularly voltage (V_e) and angle (δ_e), experience significant fluctuations when decision factors such as transformer output, line power, and battery power change. Previous research [30] supports this finding, showing that line flow is particularly sensitive to voltage changes. The electrical network exhibits the strongest coupling with

decision variables, suggesting it responds most vigorously to changes in transmission, storage, and power generation systems, as indicated by its large total sensitivity value. Therefore, it is essential to focus on improving decision factors related to electricity to minimize fluctuations and enhance the overall stability and efficiency of the system. The overall sensitivity value of the thermal network is 2.062236333. This value indicates that the thermal network is less susceptible to changes in its decision factors, especially when compared to the electrical network, which has a much higher sensitivity value. The total sensitivity value of 2.062236333 suggests that the thermal network experiences a reduced overall impact, even in the presence of significant variables affecting gas pressure and temperature, such as boiler thermal output ($H_{\text{Boiler}}^{\text{g}}$) and CHP thermal output ($H_{\text{chp-heat}}^{\text{h}}$). This lower sensitivity may imply that while thermal decisions are important, the thermal network does not respond as strongly to operational modifications as the electrical network does. Although it remains essential to monitor and optimize thermal variables, their overall impact on the system is less pronounced. The water network exhibits the lowest overall sensitivity value among the three networks, measuring at 0.0352135. This low sensitivity indicates that changes in decision variables, such as water flow or desalination, have minimal impact on the water network. Although the water network is essential for the overall operation of the system, it responds very little to fluctuations in decision factors. This can be attributed to the comparatively low sensitivity values associated with water flow-related decision variables, such as desalination ($W^{\text{desalination}}$) and water flow through pipes ($W^{\text{wat-line}}$), which are significantly lower than those observed in the thermal and electrical networks. In conclusion, the electricity network remains the most susceptible to changes in decision variables, despite the importance of the thermal and water networks. This situation necessitates strategic planning and targeted interventions to ensure efficient operations and minimize costs within the overall energy hub system.

Jacobian Matrix for Electrical Network under FDIA

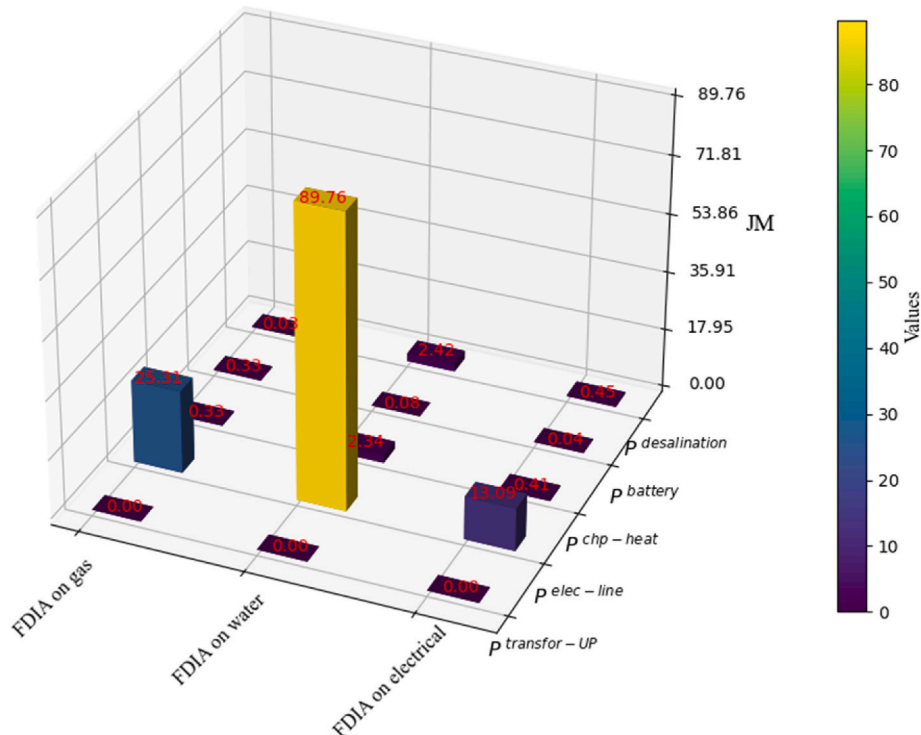


Fig. 4. The Jacobian matrix of the electrical network based on three attacks for the change in 25 % of data within network.

Jacobian Matrix for Thermal Network under FDIA

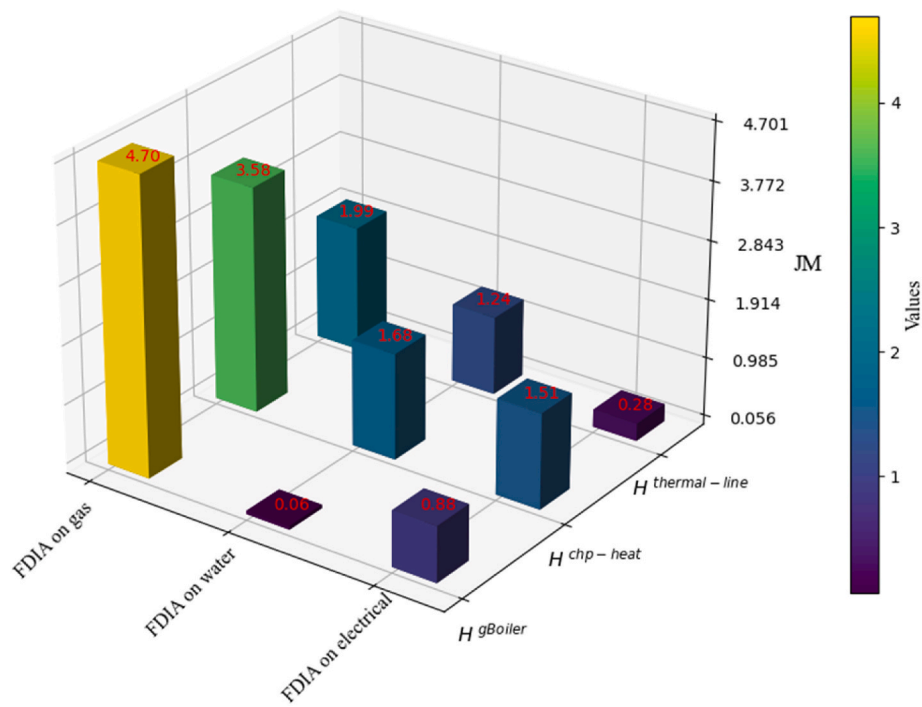


Fig. 5. The Jacobian matrix of thermal network based on three attacks for the change in 25 % of data within network.

Jacobian Matrix for Thermal Network under FDIA

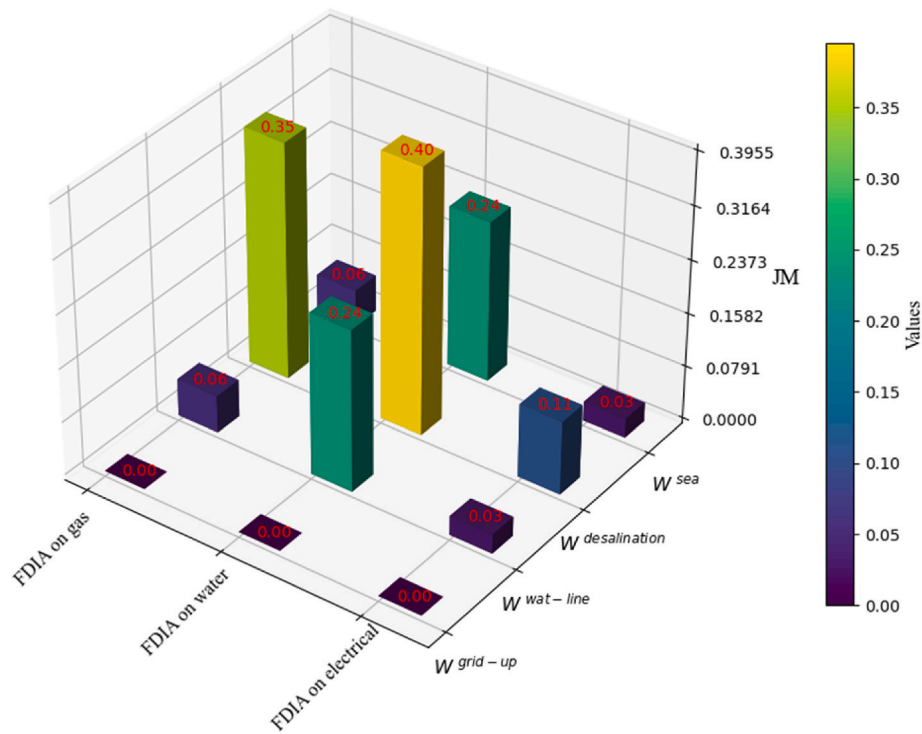


Fig. 6. The Jacobian matrix of the water network based on three attacks for the change in 25 % of data within network.

Analyzing the Jacobian matrices for the thermal, water, and electrical networks reveals different patterns of sensitivity to FDIA as indicated in Figs. 4–6. The electrical system in Fig. 4 demonstrates a high level of sensitivity to these attacks, with the greatest sensitivity value of 89.76 associated with FDIA (the change in 25 % of data within network) affecting electrical decisions, specifically the line flow. Similarly, line power shows significant sensitivity, indicating that decisions regarding electrical operations have a considerable impact on the network's behavior. In the thermal network (see Fig. 5), there is a peak sensitivity of 4.70 related to FDIA on gas that influences boiler thermal production. This indicates that the thermal network is particularly vulnerable to FDIA within the gas sector, suggesting that cyberattacks targeting the gas network especially those aimed at boiler operations could severely disrupt the thermal system. Based on Fig. 6, the water network shows the highest sensitivity to cyberattacks, particularly regarding decisions related to and desalination, with sensitivity values of 0.40. These findings underscore the importance of directing cybersecurity efforts toward the most vulnerable components of each network. For instance, the water network must ensure the safe flow of water and the efficiency of desalination operations. Similarly, the thermal network should focus on safeguarding gas-to-thermal conversion processes, while the electrical network must prioritize the protection of transformers and power lines. This vulnerability across all networks highlights the need for targeted security measures to mitigate the impact of cyberattacks.

2) The comparative evaluation of proposed risk management (PJM) in IWPS

The power load and production across the electrical, thermal, and water networks over a 24-h period are illustrated in Figs. 7–9. The electrical network load varies significantly, peaking at approximately 5000 kW, indicating substantial demand during certain hours. The primary sources of electricity production are the upstream grid and injected power of lines, which contribute significantly, especially during peak demand times. In contrast, desalination and batteries contribute less to overall power generation and show less variability. This suggests that while transformers and line outputs dominate, batteries and desalination provide supplemental power generation. Boiler thermal and CHP thermal are the primary sources of power generation in the thermal network. These sources experience significant increases in power production between the hours of 10 a.m. and 3 p.m., aligning with heightened thermal demand during this period. The thermal line highlights its role in providing consistent thermal support by maintaining a constant, though reduced, output throughout the day. The graph

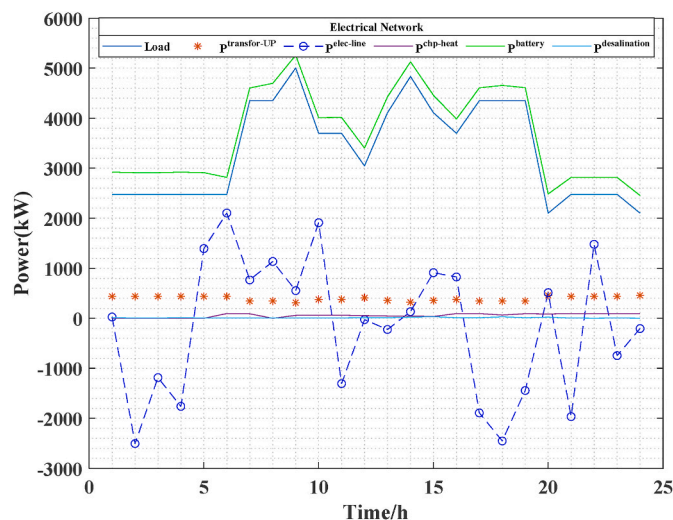


Fig. 7. The load and output generations of electrical network.

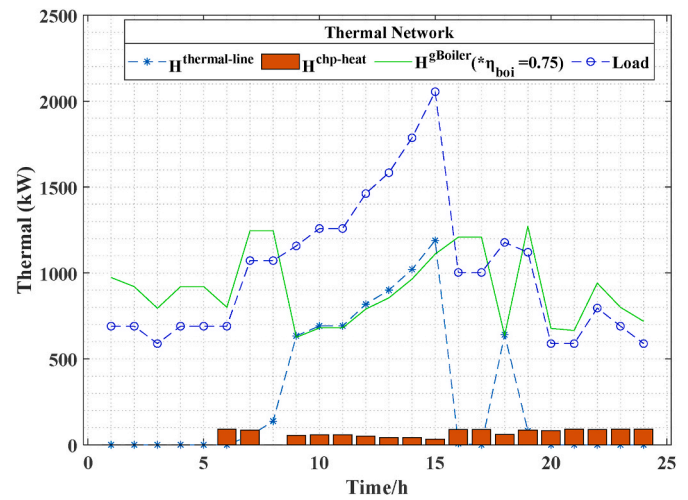


Fig. 8. The load and output generations of thermal network.

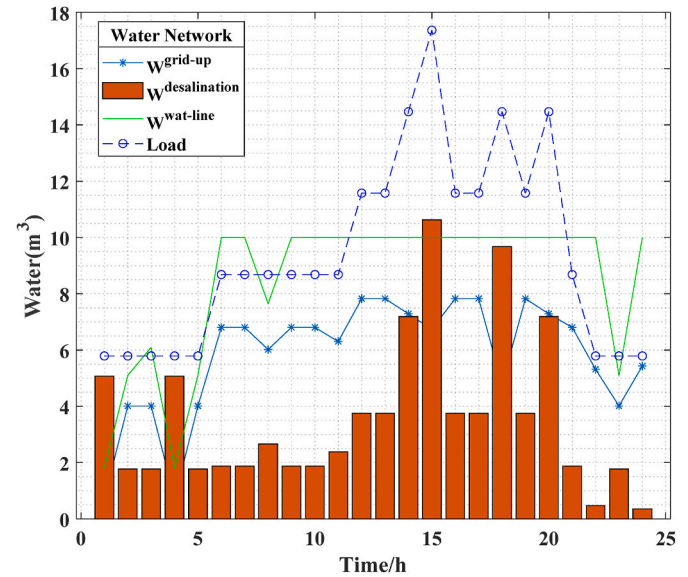


Fig. 9. The load and output generations of the water network.

illustrates how boiler and CHP systems are utilized to meet peak thermal energy demands, particularly during energy recovery or industrial heating processes. In contrast, power output and consumption in the water network show more variability, especially due to desalination and water-line activities. There are notable peaks in the desalination process between the hours of 10 a.m. and 3 p.m., indicating a substantial demand for water processing during this time. While power consumption from the grid remains relatively stable, the water line shows significant fluctuations throughout the day, indicating changing water flow requirements. The load on the water network is fairly constant, with occasional peaks that align with the demands of desalination and water distribution. This variability suggests that fluctuations in the need for water processing and distribution potentially influenced by external factors such as supply conditions or consumption trends are closely linked to the water network's power consumption. In conclusion, there are distinct differences in the power consumption patterns among the three networks, with the electrical network being the most variable due to the operations of transformers and lines. The water network shows fluctuating power usage primarily due to desalination and water flow processes. In contrast, the thermal network relies on boilers and CHP systems to meet peak heating demands. To maintain system stability and

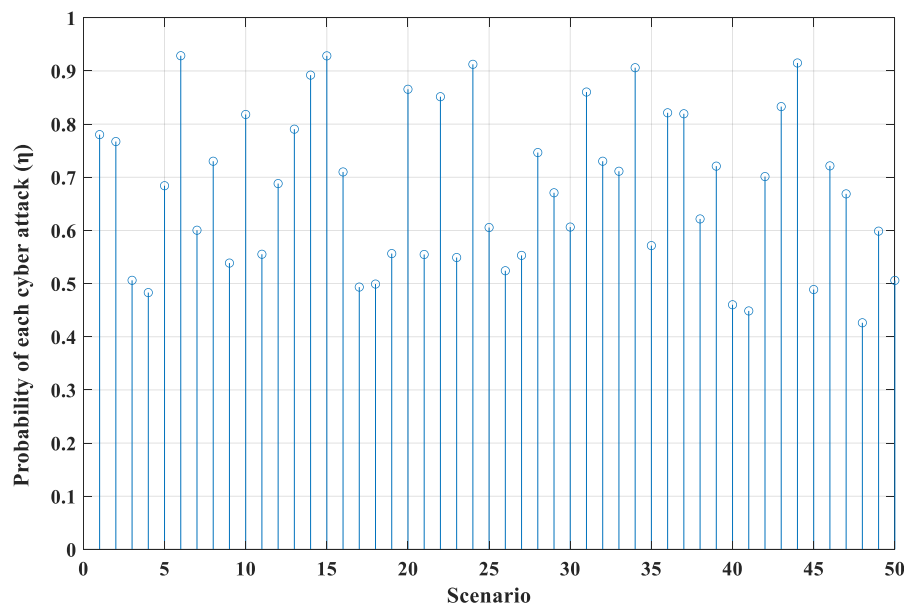


Fig. 10. The probability coefficient for each scenario (the change in 25 % of data).

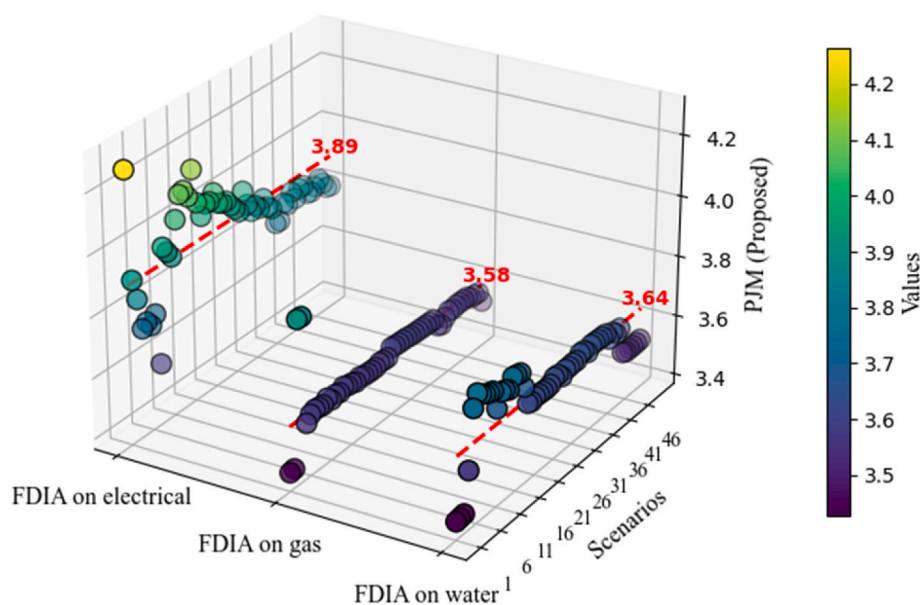


Fig. 11. The risk assessment of IWPS for 50 scenarios.

minimize inefficiencies throughout the 24-h cycle, it is crucial to manage these varying loads and power generation sources effectively.

The consequences of 50 distinct scenarios simulating FDIAs on gas, water, and electrical networks are illustrated in Figs. 10–12. These scenarios were evaluated from two perspectives: total cost and PJM risk. the probability coefficient for each scenario is illustrated in Fig. 10. Each point on Fig. 11 represents a scenario involving an FDIA on one of the networks, plotted according to its PJM values. The FDIA scenario affecting the electrical network, specifically the sixth scenario, has the highest color-coded PJM value of 3.89, which falls within a range of 3.5–4.2. Higher PJM values indicate that FDIAs targeting the electrical network are associated with the greatest risk. The FDIA values for gas indicate a lower but still significant sensitivity, primarily around 3.6 in various network scenarios. In contrast, the FDIA values for water scenarios show PJM values ranging between 3.58 and 3.64, suggesting a reasonably substantial risk. The overall costs associated with these

scenarios, which range from 0 to 1.6×10^{13} , are illustrated in Fig. 12. Notably, the 46th case presents the highest FDIA cost for the electrical network scenario at 1×10^{13} , highlighting that cyberattacks on the electrical system incur significantly higher expenses. In comparison, the lower-cost scenarios primarily involve FDIA on gas and water, where the overall costs are substantially lower, typically ranging between 1×10^{11} and 1×10^{12} . The significant impact of electrical network attacks on the overall system is highlighted by the red marker in both graphs, which indicates the maximum value. Additionally, there is a stronger link between FDIA in the electrical network and the associated risks and costs, as shown by the average values for both the PJM and overall costs. Compared to gas and water networks, the electrical network is the most vulnerable to FDIA, resulting in higher PJM values and increased overall costs, as demonstrated by the analysis. These findings emphasize the critical importance of protecting the electrical network from cyberattacks, given the greatest risk and potential for financial loss. On the

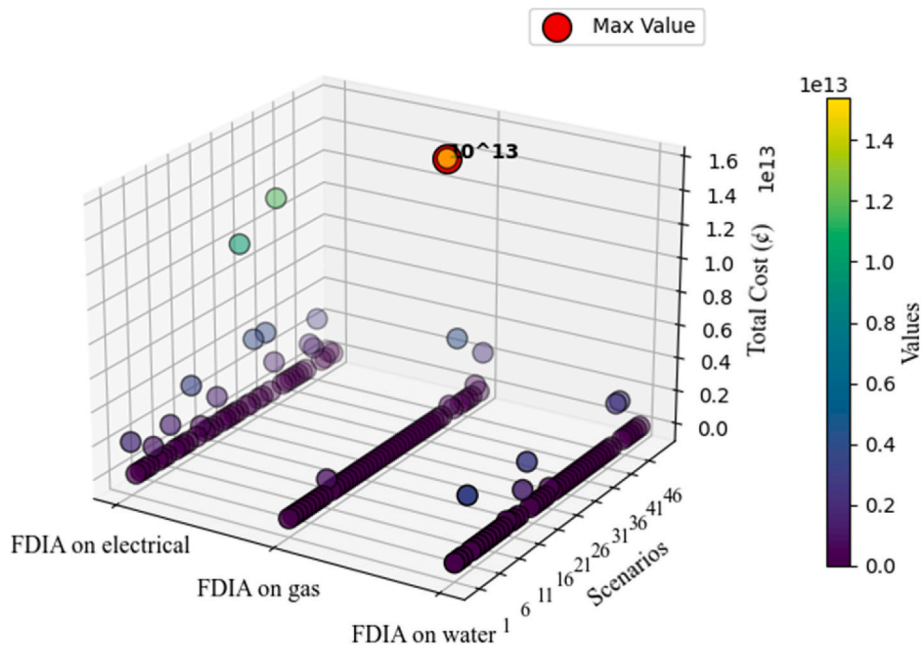


Fig. 12. The total cost of IWPS for 50 scenarios.

other hand, while FDIA can disrupt gas and water networks, the associated risks and financial consequences are significantly lower. To minimize risks and costs in any scenario, it is essential to prioritize the

security of the electrical network. At the same time, adequate safeguards should be implemented to protect the gas and water systems.

The advantages of the PJM technique become apparent when

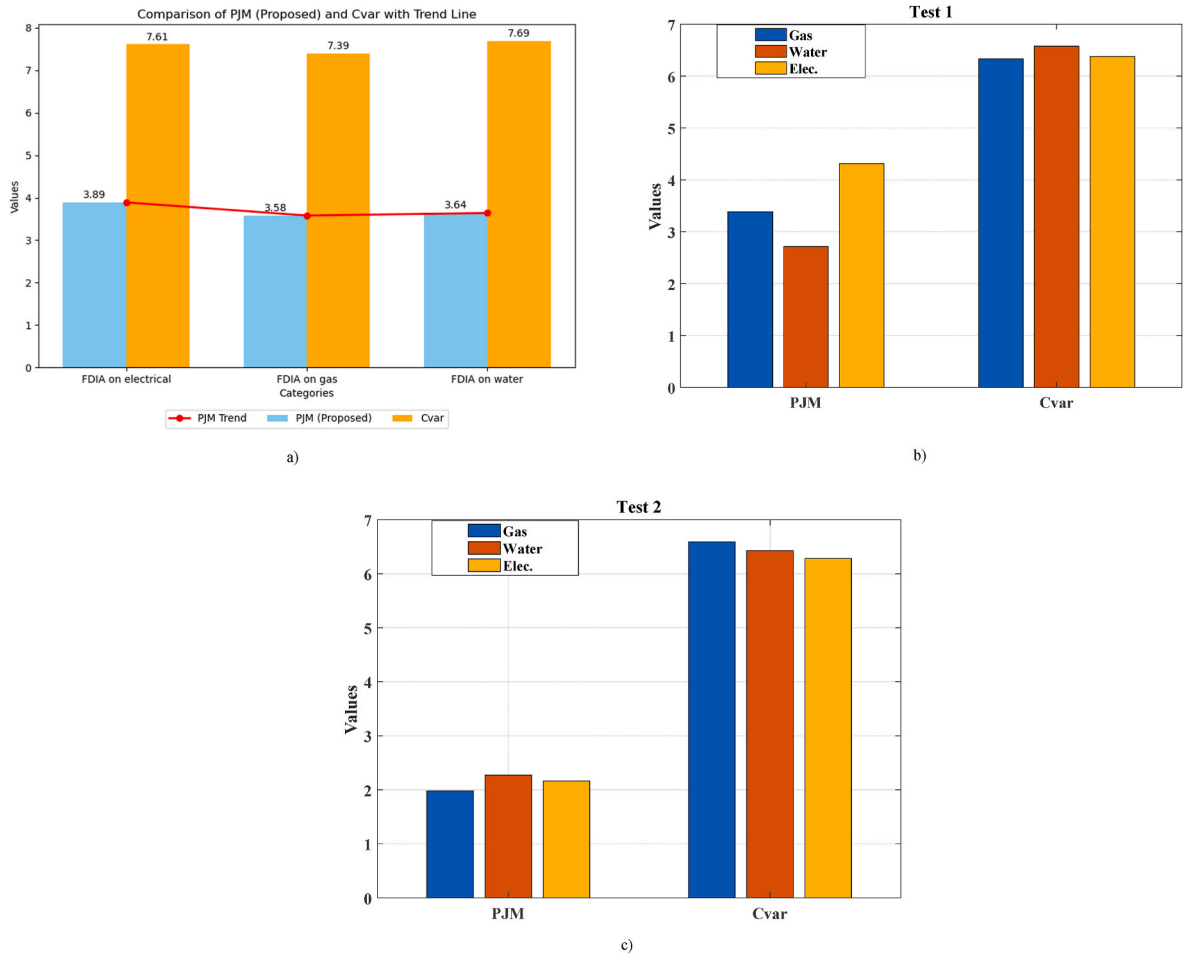


Fig. 13. The comparative result of risk assessment based on PJM and CVaR: a) total risk assessment b) Test 1 c) Test 2.

compared to the Conditional Value-at-Risk (CVaR) method in assessing the risks of False Data Injection Attacks (FDIA) across gas, water, and electrical networks. While CVaR is commonly used to evaluate tail-end risks, it often leads to overestimations due to its tendency to amplify extreme outcomes—especially in multi-carrier energy systems with dynamic interdependencies. In contrast, PJM offers a more stable and cautious representation of risk, aligning better with operational realities. When comparing the PJM and the Conditional Value-at-Risk (CVar) approaches, it is evident that PJM provides more reliable and accurate risk estimations for FDIA in gas, water, and electricity networks. For the electrical network in Fig. 13, CVar reports an FDIA value of 7.61, while PJM offers a lower value of 3.64, indicating that PJM provides a more trustworthy and conservative risk forecast. Similarly, for the gas network, PJM has a risk value of 3.89 compared to CVar's 7.39, showing that PJM consistently yields lower and more accurate results. This trend is also reflected in the water network, where CVar is at 7.69 and PJM is at 3.58. Although Cvar shows greater variability, the PJM values remain relatively close to each other, indicating stability. In contrast to the significant fluctuations seen in Cvar, the trend line for PJM reveals a steady and consistent method of risk estimation. This makes the PJM approach a more effective tool for risk assessment in such situations, as it tends to be more accurate and less prone to over-estimation. Notably, PJM demonstrates consistent performance across the three networks, with its estimated risk values remaining within a relatively narrow range. This stability indicates that PJM's modeling approach is robust, making it more suitable for protecting critical infrastructure in unpredictable situations. Conversely, the greater variation in CVaR results highlights its susceptibility to outliers and model assumptions, which could pose challenges in real-time decision-making. Overall, this comparative analysis underscores the effectiveness of PJM as a practical method for estimating cyber risk in integrated energy systems, where operational accuracy and resilience are vital. We conducted two testing phases to validate the proposed risk assessment method, as shown in Fig. 13.

3) The robustness analysis of energy converters in risk management

The primary objectives of the multi-objective optimization problem related to the addition of new energy converters to the system are cost reduction and the minimization of PJM risk. The system comprises five different energy converters: a photovoltaic (M_{PV}), an electricity-to-thermal converter (M_{EH}), a water-to-thermal converter (M_{WH}), a desalination unit ($M_{desalination}$), and a combined thermal and power unit (M_{chp}). The main goal of this investigation is to identify the optimal combination of these converters that can effectively reduce both PJM risk and costs simultaneously. To evaluate the efficacy of various combinations, seven distinct cases representing different mixes of these converters have been considered. Three criteria are used to evaluate each case: Cvar, PJM risk, and cost. To evaluate system performance, the proposed two-stage optimization methodology utilizes two key performance indicators. The first is the risk coefficient, which is calculated by integrating the sensitivity findings from the Proposed Jacobian Matrix (as shown in equation (49)) and is computed using equation (52). This metric provides a scalar measure of overall system vulnerability and indicates how state variables respond to disruptions caused by

cyberattacks. The second factor is the overall operational cost, determined by equation (68). This encompasses all energy-related expenses, including gas consumption, water purchases, and electricity imports. To assess improvements in cost performance, the percentage decrease in operating costs is calculated under the proposed framework compared to a baseline or traditional scenario. The situations are assessed across four distinct scenarios: FDIA in the electrical network, FDIA in the gas network, FDIA in the water network, and a combined FDIA in both the gas and electrical networks. Based on Tables 3–6 compared to the other cases, examples 5, 6, and 7 demonstrate better performance in terms of reducing PJM risk and costs. Among these, Case 6 stands out as the most efficient overall, offering the best converter combination to minimize both risk and expenses. There are no noticeable improvements in either PJM risk or Cvar in Case 1, which uses a mix of converters without any specific focus. The costs are at 2.5012×10^{13} , with PJM at 7.35644 and Cvar at 7.00139. Since neither the risk nor the cost is significantly reduced, these results indicate that the combination of converters is rather inefficient. In Case 2, there is a slight improvement in both PJM risk and Cvar values; however, it is not enough to be considered an ideal solution. In this scenario, PJM is 6.28375, Cvar is 6.591021, and the cost remains at 2.5013×10^{13} . This suggests that even though the design of the converters has been somewhat modified, it does not lead to an efficient reduction in both cost and risk. Although Case 3 provides additional evidence of optimization, it is still insufficient to significantly impact PJM and Cvar. In this case, the cost is 2.425×10^{13} , with PJM at 6.378301 and Cvar at 6.33117. While these values are slightly better than those in the first two cases, they remain quite high. This indicates that the converter combination used in Case 3 does not effectively optimize the system for reducing risk and costs. In Case 4, further changes are introduced, but the improvements are minimal. The cost is 2.378×10^{13} , with PJM at 6.314279 and Cvar remaining at 6.378301. The decreases in PJM and Cvar are not significant enough to suggest that this combination is the most effective. This instance reinforces the idea that simply increasing the number of converters does not necessarily lead to a substantial reduction in risk or costs. In Case 5, a more targeted combination of converters leads to a noticeable improvement in both PJM and Cvar metrics. The cost decreases significantly to 9.7507×10^7 , with PJM values at 2.34747 and Cvar at 6.759645. This reflects a clear reduction in both cost and PJM risk compared to previous cases. However, while Case 5 shows meaningful gains, it is not the best combination when compared to Cases 6 and 7. Case 6 demonstrates even greater progress, with the cost further reduced to 9.7508×10^7 . In this case, PJM drops to 2.378030, and Cvar decreases to 6.830607. The findings from Case 6 indicate that the most efficient and cost-effective option minimizes both PJM risk and costs by thoughtfully integrating all energy converters.

The analysis of the Particle Swarm Optimization (PSO) technique for determining the optimal configuration of energy converters in the system is illustrated in Figs. 14 and 15. Fig. 14 displays the distribution of the PSO population data, highlighting the dispersion of PJM values across various groups and scenarios. The average grid value is marked by red points on the graph, which indicate the lowest PJM value at 2.59 and the highest at 3.98. These values illustrate the variability of PJM among different converter combinations within the PSO population. The range of PJM values, from 2.59 to 3.98, reflects the differences in risk

Table 3
The comparison of different studied cases for FDIA on gas network.

Scenarios	FDIA on the gas network						
	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
$M_{PV}, M_{WH}, M_{EH}, M_{desalination}, M_{chp}$	1-0-0-0-0	0-0-0-0-0	1-0-0-0-1	0-0-0-1-1	0-1-1-1-0	1-1-1-1-1	1-1-1-0-0
Cost (\$)	9.75241×10^7	9.7529×10^7	9.75293×10^7	9.7529×10^7	9.7507×10^7	9.7507×10^7	9.7507×10^7
PJM (Proposed)	2.03341	1.981842	2.31670	3.38510	2.04368	2.52714	2.65697
Cvar [19]	7.8503	6.591021	7.69691	6.33117	7.0378	7.0229	6.9608

Table 4

The comparison of different studied cases for FDIA on water network.

Scenarios	FDIA on the water network						
	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
$M_{PV}, M_{WH}, M_{EH},$ $M_{desalination}, M_{chp}$	1-0-0-0-0	0-0-0-0-0	1-0-0-0-1	0-0-0-1-1	0-1-1-1-0	1-1-1-1-1	1-1-1-0-0
Cost (\$)	9.75242×10^7	9.7525×10^7	9.7523×10^7	9.75243×10^7	9.7507×10^7	9.7507×10^7	9.7507×10^7
PJM (Proposed)	2.932587	2.27574	2.27819	2.71447	2.00003	2.38054	2.0005
Cvar [19]	7.74097	6.427167	7.455307	6.57442	6.6375	6.93386	6.806160

Table 5

The comparison of different studied cases for FDIA on the electrical network.

Scenarios	FDIA on the electrical network						
	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
$M_{PV}, M_{WH}, M_{EH},$ $M_{desalination}, M_{chp}$	1-0-0-0-0	0-0-0-0-0	1-0-0-0-1	0-0-0-1-1	0-1-1-1-0	1-1-1-1-1	1-1-1-0-0
Cost (\$)	2.5012×10^{13}	2.5013×10^{13}	2.378×10^{13}	2.378×10^{13}	9.7507×10^7	9.7508×10^7	9.7508×10^7
PJM (Proposed)	2	2.16639	2.451095	4.314279	2.34747	2.378030	2.32870
Cvar [19]	7.356440	6.28375	7.00139	6.378301	6.759645	6.830607	6.92316

Table 6

The comparison of different studied cases for FDIA on gas and electrical networks.

Scenarios	Coordinated FDIA on (gas and electrical)						
	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
$M_{PV}, M_{WH}, M_{EH},$ $M_{desalination}, M_{chp}$	1-0-0-0-0	0-0-0-0-0	1-0-0-0-1	0-0-0-1-1	0-1-1-1-0	1-1-1-1-1	1-1-1-0-0
Cost (\$)	2.50124×10^{13}	2.507×10^{13}	2.425×10^{13}	2.484×10^{13}	9.7507×10^7	9.75081×10^7	9.7508×10^7
PJM (Proposed)	2.24799	2.34848	3.44016	5.450696	2.741670	3.28074	2.56161
Cvar [19]	7.58346	4.7465	7.880210	6.602708	6.897147	6.96582	6.87235

across the various scenarios. The plot indicates that there may be opportunities for improvement in certain situations. However, the distribution of PJM values shows a range of high peaks and low valleys, suggesting that PJM risk is being optimized within a specific range. The Particle Swarm Optimization (PSO) algorithm was specifically designed to address a bi-objective problem that involves the simultaneous minimization of PJM-based cyber risk (referred to as objective function F1) and total operational cost (designated as F2). The optimization results are illustrated in Fig. 15C. This figure depicts the trade-off surface between these two competing objectives, demonstrating how different energy system topologies affect risk exposure and economic efficiency. In Case 11111, all five energy converters in the system are operational. At this stage, the total cost (F2) is limited to 9.75×10^7 monetary units, while the PJM-based risk (F1) is minimized to a value of 3.17. This outcome indicates that Case 11111 represents the optimal mix of converters within the proposed framework, as it achieves the best balance between system security and economic performance. The convergence in this direction illustrates how effectively the PSO binary two-objective optimization algorithm explores the solution space to identify configurations that minimize risk at the lowest possible cost. The PSO algorithm's consistent ability to maintain this balance emphasizes its potential utility in actual energy system design, especially in scenarios where careful management of cost and resilience trade-offs is essential. Additionally, the relatively smooth appearance of the Pareto front in Fig. 15 indicates that even small increases in risk or cost typically require significant trade-offs in the other metric. This further supports the selection of Case 11111 as a robust and balanced solution that aligns well with the goals of designing an integrated energy system while adhering to cyber-security regulations. The optimization findings reveal that Case 11111 is the most effective configuration, successfully addressing the multi-objective optimization problem by minimizing both cost and risk (PJM). These outcomes highlight the effectiveness of

the PSO method in tackling complex optimization challenges by managing multiple goals simultaneously and delivering a well-rounded, optimal solution.

7. Conclusion

This paper carefully examined the sensitivity, risks, and financial implications of various factors and cyberattacks on the thermal, electrical, and water networks within an IWPS. Hence, this study proposed an efficient risk analysis for IWPSs with emphasis on the Jacobean based sensitivity function among three electricity, thermal, and water networks. Then, a two-objective strategy was proposed to reduce the risk value and total cost using the new energy converters in IWPS. To do so, we implemented the proposed framework on a studied IWPS and analyzed the relative results. It is deduced from the results that the electricity network is particularly important and is the most sensitive to these factors, as demonstrated by a transformer output sensitivity value of 161.6068, which significantly affects both voltage (V_e) and angle (δ_e). Additionally, the study identified FDIA as the most significant threat to the electrical network, compared to the gas and water networks, with a maximum economic impact of 1.6×10^{13} and the highest PJM risk value of 3.89. The stability and accuracy of the PJM approach were demonstrated through comparative risk analysis of the PJM and the Cvar methods. The risks associated with potential FDIA for the water, gas, and electrical networks were calculated to be 3.64, 3.89, and 3.58, respectively. These values were significantly lower and more reliable than the Cvar values, which ranged from 7.39 to 7.69. In an IWPS, optimizing energy converters using the PSO method has proven to be an effective strategy for reducing both PJM risk and costs. The best results were achieved in Case 6, which corresponds to the configuration of three energy converters including PV, water-to-thermal, and electricity-to-thermal, based on an analysis of seven potential case combinations. In

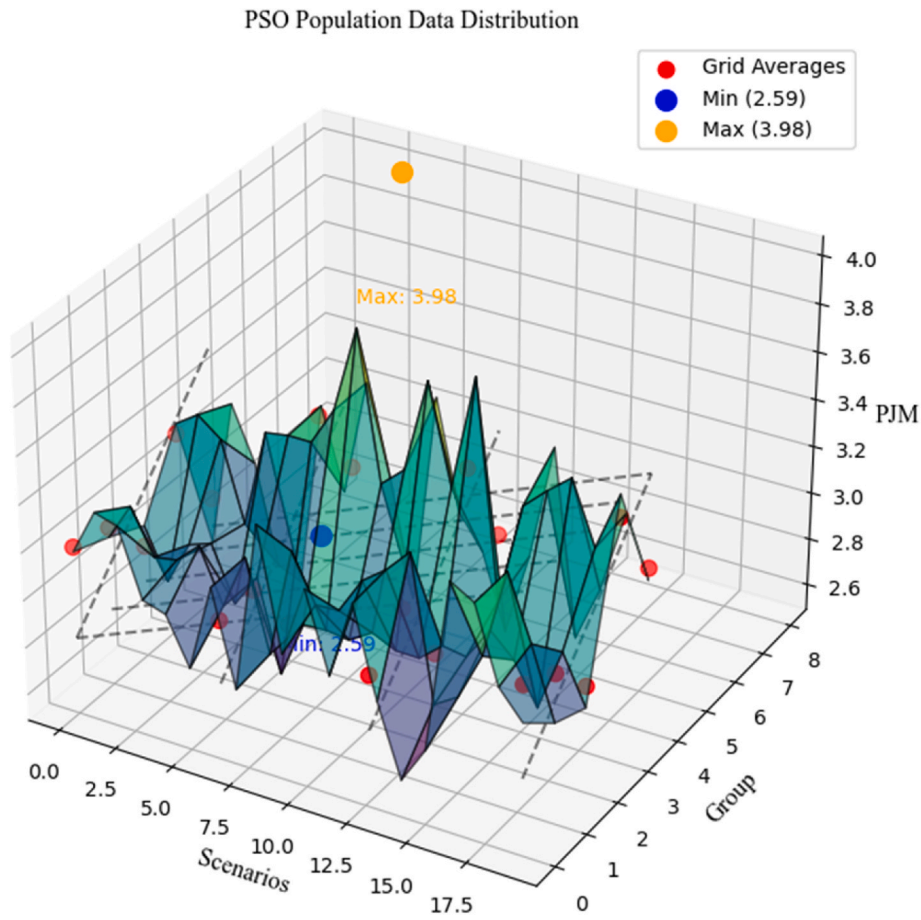


Fig. 14. The population data distribution of the PSO algorithm.

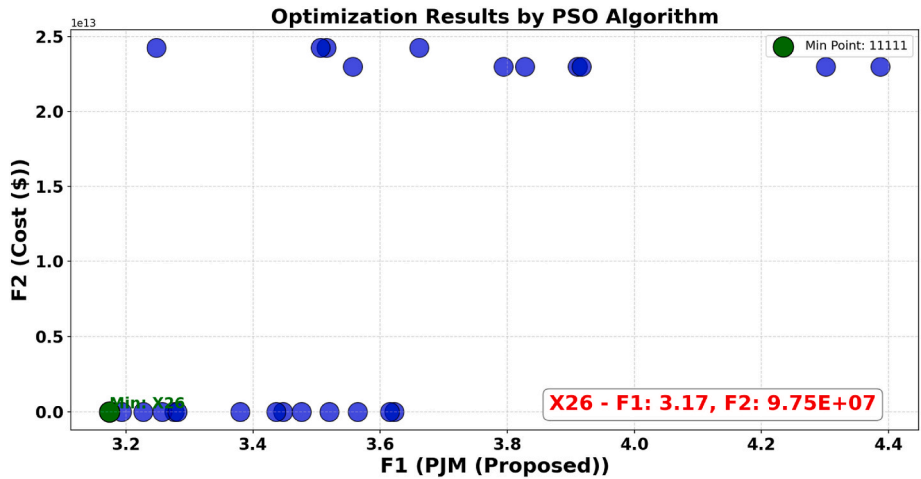


Fig. 15. The optimization results of the PSO algorithm.

this case, the cost was significantly lowered to 9.7507×10^7 , and the PJM was reduced to 2.32870. The PJM risk levels ranged from a minimum of 2.59 to a maximum of 3.98, demonstrating the PSO algorithm's effectiveness in navigating the solution space and identifying optimal configurations. With F1 (PJM) at 3.17 and F2 (cost) at 9.75×10^7 , Case 6 struck the optimal balance between minimizing risk and expense, based on the PJM values derived from the PSO optimization findings. Overall, the study illustrates the effectiveness of PSO in addressing multi-objective optimization challenges, with Case 6 providing the best

combination of cost savings and risk reduction. In future work, we aim to enhance our current two-stage approach by integrating stochastic optimization models, such as robust mixed-integer linear programming (MILP) or chance-constrained programming.

CRediT authorship contribution statement

Hamzeh Roghanian Jahromi: Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology,

Investigation, Formal analysis, Data curation, Conceptualization. **Mohsen Gitizadeh**: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Formal analysis, Data curation, Conceptualization. **Morteza Sheikh**: Writing – review & editing, Writing – original draft, Software, Resources, Investigation, Conceptualization. **Ali Akbar Ghasemi**: Writing – review & editing, Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

APPENDIX

The Jacobian matrix for the IWPS comprises 12 control variables (rows) and 5 state variables (columns), capturing the sensitivity of control outputs to state variations as shown in (77). Key state variables include $V_e, \delta_e, P_{gas-total}^S, T_{RL}^S, h_{ww,t}$. This matrix enables systematic analysis of control-state interactions, critical for predicting system behavior under disturbances such as cyberattacks or unexpected oscillations. By quantifying these relationships, operators enhance control robustness and resilience, ensuring stability in dynamic operational scenarios. The Jacobian thus serves as a foundational tool for system diagnostics and adaptive control design in complex power systems.

$$\begin{aligned}
 \text{JM} = & \begin{array}{c} \left. \begin{array}{l} \text{Electricity Network} \\ \text{Thermal Network} \end{array} \right\} \begin{array}{cc} \begin{array}{c} \frac{\partial P_{transfor-UP}}{\partial V_e} \\ \frac{\partial P_{elec-line}}{\partial V_e} \\ \frac{\partial P_{battery}}{\partial V_e} \\ \frac{\partial P_{desalination}}{\partial V_e} \\ \frac{\partial P_{chp-heat}}{\partial V_e} \\ \frac{\partial H_{chp-heat}}{\partial V_e} \\ \frac{\partial H_{gBoiler}}{\partial V_e} \\ \frac{\partial H_{thermal-line}}{\partial V_e} \\ \frac{\partial W_{grid-up}}{\partial V_e} \\ \frac{\partial W_{desalination}}{\partial V_e} \\ \frac{\partial W_{wat-line}}{\partial V_e} \\ \frac{\partial W_{sea}}{\partial V_e} \\ \frac{\partial W_{ww' \leq ww,t}}{\partial V_e} \end{array} & \begin{array}{c} \frac{\partial P_{transfor-UP}}{\partial \delta_e} \\ \frac{\partial P_{elec-line}}{\partial \delta_e} \\ \frac{\partial P_{battery}}{\partial \delta_e} \\ \frac{\partial P_{desalination}}{\partial \delta_e} \\ \frac{\partial P_{chp-heat}}{\partial \delta_e} \\ \frac{\partial H_{chp-heat}}{\partial \delta_e} \\ \frac{\partial H_{gBoiler}}{\partial \delta_e} \\ \frac{\partial H_{thermal-line}}{\partial \delta_e} \\ \frac{\partial W_{grid-up}}{\partial \delta_e} \\ \frac{\partial W_{desalination}}{\partial \delta_e} \\ \frac{\partial W_{wat-line}}{\partial \delta_e} \\ \frac{\partial W_{sea}}{\partial \delta_e} \\ \frac{\partial W_{ww' \leq ww,t}}{\partial \delta_e} \end{array} & \begin{array}{c} \frac{\partial P_{transfor-UP}}{\partial P_{gas-total}} \\ \frac{\partial P_{elec-line}}{\partial P_{gas-total}} \\ \frac{\partial P_{battery}}{\partial P_{gas-total}} \\ \frac{\partial P_{desalination}}{\partial P_{gas-total}} \\ \frac{\partial P_{chp-heat}}{\partial P_{gas-total}} \\ \frac{\partial H_{chp-heat}}{\partial P_{gas-total}} \\ \frac{\partial H_{gBoiler}}{\partial P_{gas-total}} \\ \frac{\partial H_{thermal-line}}{\partial P_{gas-total}} \\ \frac{\partial W_{grid-up}}{\partial P_{gas-total}} \\ \frac{\partial W_{desalination}}{\partial P_{gas-total}} \\ \frac{\partial W_{wat-line}}{\partial P_{gas-total}} \\ \frac{\partial W_{sea}}{\partial P_{gas-total}} \\ \frac{\partial W_{ww' \leq ww,t}}{\partial P_{gas-total}} \end{array} & \begin{array}{c} \frac{\partial P_{transfor-UP}}{\partial T_{RL}^S} \\ \frac{\partial P_{elec-line}}{\partial T_{RL}^S} \\ \frac{\partial P_{battery}}{\partial T_{RL}^S} \\ \frac{\partial P_{desalination}}{\partial T_{RL}^S} \\ \frac{\partial P_{chp-heat}}{\partial T_{RL}^S} \\ \frac{\partial H_{chp-heat}}{\partial T_{RL}^S} \\ \frac{\partial H_{gBoiler}}{\partial T_{RL}^S} \\ \frac{\partial H_{thermal-line}}{\partial T_{RL}^S} \\ \frac{\partial W_{grid-up}}{\partial T_{RL}^S} \\ \frac{\partial W_{desalination}}{\partial T_{RL}^S} \\ \frac{\partial W_{wat-line}}{\partial T_{RL}^S} \\ \frac{\partial W_{sea}}{\partial T_{RL}^S} \\ \frac{\partial W_{ww' \leq ww,t}}{\partial T_{RL}^S} \end{array} & \begin{array}{c} \frac{\partial P_{transfor-UP}}{\partial h_{ww,t}} \\ \frac{\partial P_{elec-line}}{\partial h_{ww,t}} \\ \frac{\partial P_{battery}}{\partial h_{ww,t}} \\ \frac{\partial P_{desalination}}{\partial h_{ww,t}} \\ \frac{\partial P_{chp-heat}}{\partial h_{ww,t}} \\ \frac{\partial H_{chp-heat}}{\partial h_{ww,t}} \\ \frac{\partial H_{gBoiler}}{\partial h_{ww,t}} \\ \frac{\partial H_{thermal-line}}{\partial h_{ww,t}} \\ \frac{\partial W_{grid-up}}{\partial h_{ww,t}} \\ \frac{\partial W_{desalination}}{\partial h_{ww,t}} \\ \frac{\partial W_{wat-line}}{\partial h_{ww,t}} \\ \frac{\partial W_{sea}}{\partial h_{ww,t}} \\ \frac{\partial W_{ww' \leq ww,t}}{\partial h_{ww,t}} \end{array} \end{array} \right\} \begin{array}{c} \text{Water Network} \end{array} \quad (77)
 \end{array}$$

$$\begin{aligned}
 & \begin{array}{cc} \begin{array}{c} 0 \\ -(2V_e g b_l - V_{e1} g b_l \cos(\delta_e) - V_{e1} b g_l \sin(\delta_e)) \\ 2V_e g b_l - V_{e1} g b_l \cos(\delta_e) - V_{e1} b g_l \sin(\delta_e) \\ 2V_e g b_l - V_{e1} g b_l \cos(\delta_e) - V_{e1} b g_l \sin(\delta_e) \\ 2V_e g b_l - V_{e1} g b_l \cos(\delta_e) - V_{e1} b g_l \sin(\delta_e) \\ \frac{K_1}{K_2} (2V_e g b_l - V_{e1} g b_l \cos(\delta_e) - V_{e1} b g_l \sin(\delta_e)) \end{array} & \begin{array}{c} 0 \\ -(V_e V_{e1} g b_l \sin(\delta_e) - V_e V_{e1} b g_l \cos(\delta_e)) \\ V_e V_{e1} g b_l \sin(\delta_e) - V_e V_{e1} b g_l \cos(\delta_e) \\ V_e V_{e1} g b_l \sin(\delta_e) - V_e V_{e1} b g_l \cos(\delta_e) \\ V_e V_{e1} g b_l \sin(\delta_e) - V_e V_{e1} b g_l \cos(\delta_e) \\ \frac{K_1}{K_2} (V_e V_{e1} g b_l \sin(\delta_e) - V_e V_{e1} b g_l \cos(\delta_e)) \end{array} & \begin{array}{c} 0 \\ \frac{C_{RL} K_2}{K_1} \\ \frac{C_{RL} K_2}{K_1} \\ \frac{C_{RL} K_2}{K_1} \\ \frac{C_{RL} K_2}{K_1} \\ -\frac{C_{RL} K_2}{K_1} \end{array} & \begin{array}{c} 0 \\ -\frac{\alpha_q \eta_{chp}}{K_1} \\ -\frac{\alpha_q \eta_{chp}}{K_1} \\ -\frac{\alpha_q \eta_{chp}}{K_1} \\ \frac{\alpha_q \eta_{chp}}{K_1} \\ \frac{\alpha_q \eta_{chp}}{K_1} \end{array} & \begin{array}{c} 0 \\ \frac{1}{2} \frac{p g}{\eta_{des} F_{wl}} (h_{ww,t} - h_{ww1,t})^{-\frac{1}{2}} \\ \frac{1}{2} \frac{p g}{\eta_{des} F_{wl}} (h_{ww,t} - h_{ww1,t})^{-\frac{1}{2}} \\ -\frac{1}{2} \frac{p g}{\eta_{des} F_{wl}} (h_{ww,t} - h_{ww1,t})^{-\frac{1}{2}} \\ \frac{1}{2} \frac{c}{\eta_{des} F_{wl}} (h_{ww,t} - h_{ww1,t})^{-\frac{1}{2}} \\ \frac{K_1}{K_2} \left(\frac{1}{2} \frac{p g}{\eta_{des} F_{wl}} (h_{ww,t} - h_{ww1,t})^{-\frac{1}{2}} \right) \end{array} \end{array}
 \end{aligned}$$

$$\begin{aligned}
 & \begin{array}{cc} \begin{array}{c} 0 \\ 0 \\ 0 \\ -\frac{\eta_{des}}{p g} (2V_e g b_l - V_{e1} g b_l \cos(\delta_e) - V_{e1} b g_l \sin(\delta_e)) \\ -\frac{\eta_{des}}{p g} (2V_e g b_l - V_{e1} g b_l \cos(\delta_e) - V_{e1} b g_l \sin(\delta_e)) \\ \frac{S}{\Delta t} \frac{\eta_{des}}{p g} (2V_e g b_l - V_{e1} g b_l \cos(\delta_e) - V_{e1} b g_l \sin(\delta_e)) \end{array} & \begin{array}{c} 0 \\ 0 \\ 0 \\ -\frac{\eta_{des}}{p g} (V_e V_{e1} g b_l \sin(\delta_e) - V_e V_{e1} b g_l \cos(\delta_e)) \\ -\frac{\eta_{des}}{p g} (V_e V_{e1} g b_l \sin(\delta_e) - V_e V_{e1} b g_l \cos(\delta_e)) \\ \frac{S}{\Delta t} \frac{\eta_{des}}{p g} (V_e V_{e1} g b_l \sin(\delta_e) - V_e V_{e1} b g_l \cos(\delta_e)) \end{array} & \begin{array}{c} -C_{RL} \\ -C_{RL} \\ C_{RL} \\ 0 \\ 0 \\ 0 \end{array} & \begin{array}{c} \frac{\eta_{chp}}{K_3} \\ \frac{C_{RL} C_p \Delta T_{gas}}{C_{fuel}} \\ 0 \\ -\frac{\eta_{des} \alpha_q \eta_{chp}}{p g K_1} \\ \frac{\eta_{des} \alpha_q \eta_{chp}}{p g K_1} \\ \frac{S}{\Delta t} \frac{\eta_{des}}{p g} \left(\frac{\alpha_q \eta_{chp}}{K_1} \right) \end{array} & \begin{array}{c} 0 \\ 0 \\ 0 \\ -\frac{1}{2 F_{wl}} (h_{ww,t} - h_{ww1,t})^{-\frac{1}{2}} \\ \frac{1}{2 F_{wl}} (h_{ww,t} - h_{ww1,t})^{-\frac{1}{2}} \\ \frac{S}{\Delta t} \left(\frac{1}{2 F_{wl}} (h_{ww,t} - h_{ww1,t})^{-\frac{1}{2}} \right) \end{array} \end{array} \quad (78)
 \end{aligned}$$

Data availability

Data will be made available on request.

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